

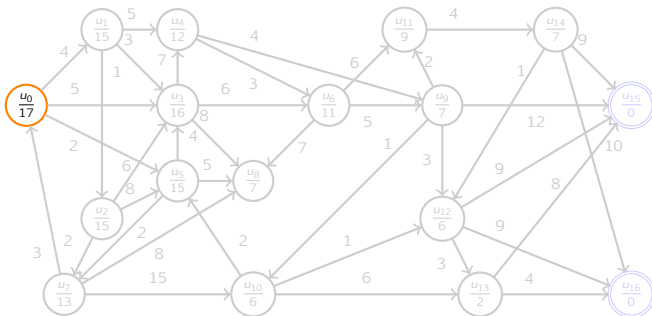
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

$u(g(u), f(u))$	père(u)	$G = \{ v \mid v(g(v), f(v)) \}$

v ← D ←



Algorithme A*(u_0, Sol)

$G \leftarrow \{u_0\}$; $D \leftarrow \emptyset$; $g(u_0) \leftarrow 0$; $f(u_0) \leftarrow 0$; $u \leftarrow \text{premier}(G)$;

while ($G \neq \emptyset$) **et** ($u \notin T$) **do**

$G \leftarrow G \setminus \{u\}$; $D \leftarrow D \cup \{u\}$; $v \leftarrow \text{suivant}(u)$;

while (v est valide) **do**

if ($(v \notin (D \cup G))$ **ou** ($g(v) > g(u) + k(u,v)$)) **then**

$g(v) \leftarrow g(u) + k(u,v)$; $f(v) \leftarrow g(v) + h(v)$;

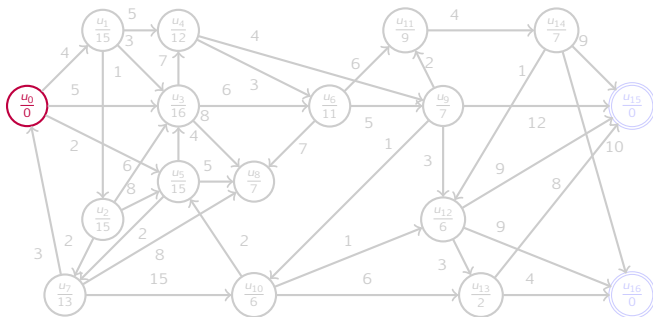
$\text{père}(v) \leftarrow u$; insérer-selon- $f(G,v)$;

$v \leftarrow \text{suivant}(u)$;

$u \leftarrow \text{premier}(G)$;

$u(g(u), f(u))$	$\text{père}(u)$	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_0(0,0)$

$v \leftarrow D = \{\}$



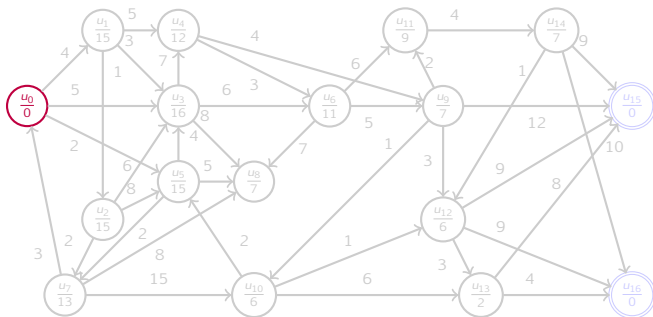
Algorithme A*(u_0, Sol)

```

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```

$u(g(u), f(u))$	père(u)	$G = \{ v \mid v(g(v), f(v)) \}$
$u_0(0,0)$	$\perp$	$u_0(0,0)$

v ← D = {}



Algorithme A*(u₀, Sol)

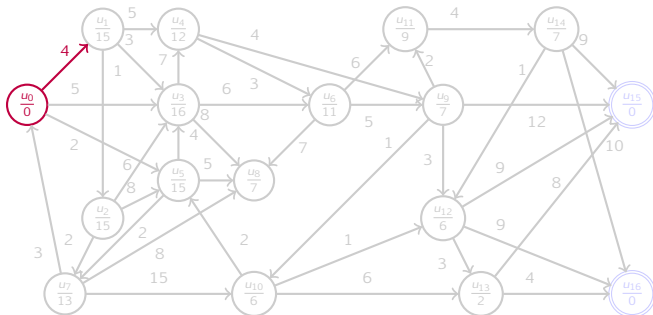
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    u ← premier(G);

```

$u(g(u), f(u))$	père(u)	$G = \{ v \mid v(g(v), f(v)) \}$
$u_0(0,0)$	$\perp$	

$v = u_1$ $D = \{u_0\}$



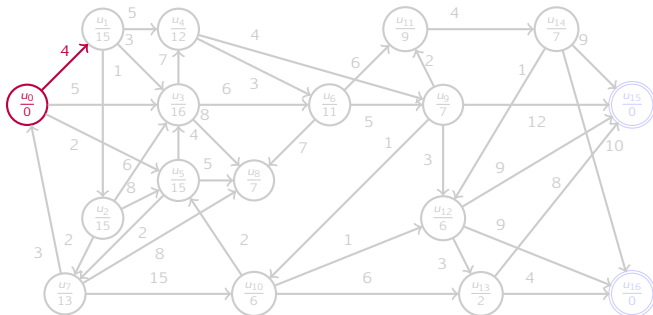
Algorithme A*(u_0, Sol)

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$u_0(0,0)$	$\perp$	

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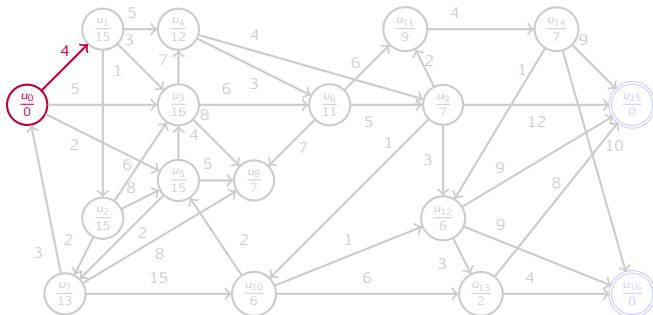
Algorithm A*(u_0, Sol)

```

G ← { $u_0$ }; D ← ∅; g( $u_0$ ) ← 0; f( $u_0$ ) ← 0; u ← premier(G);
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$u(g(u), f(u))$	père(u)	$G = \{ v \mid v(g(v), f(v)) \}$
$u_0(0,0)$	$\perp$	

$v = u_1$ D={ u_0 }



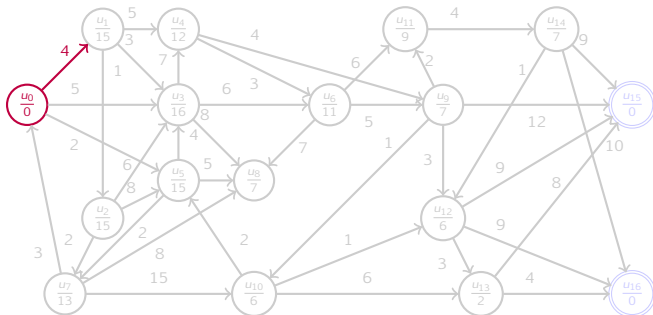
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$u(g(u), f(u))$	père(u)	$G = \{ v \mid v(g(v), f(v)) \}$
$u_0(0,0)$	$\perp$	$u_1(4,19)$

$v = u_1$ $D = \{u_0\}$



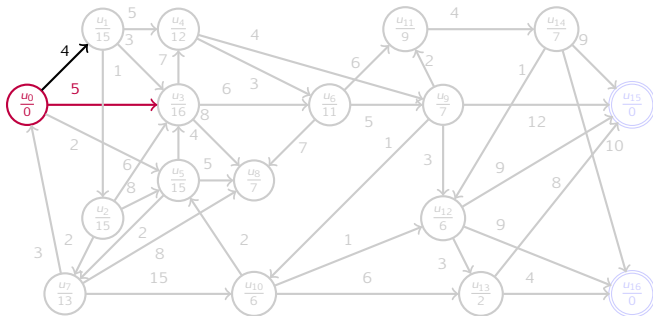
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$u_0(0,0)$	$\perp$	$u_1(4,19)$

$v = u_3$ $D = \{u_0\}$



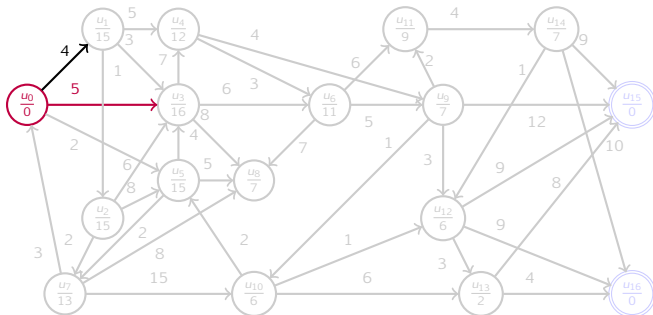
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$v = u_3$ $D = \{u_0\}$



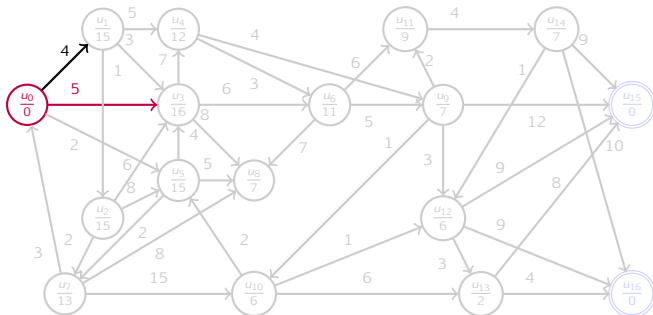
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$u_0(0,0)$	$\perp$	$u_1(4,19)$

$v = u_3$ $D = \{u_0\}$



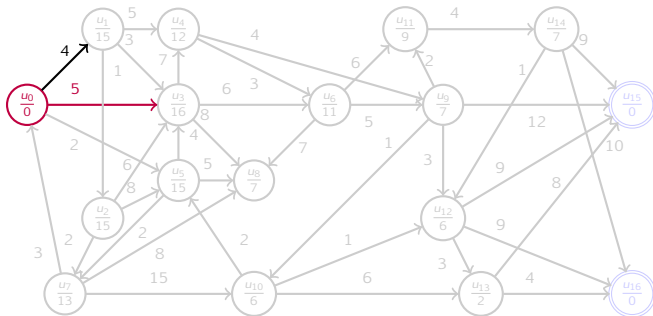
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$u_0(0,0)$	$\perp$	$u_1(4,19)$ $u_3(5,21)$

$v = u_3$ $D = \{u_0\}$



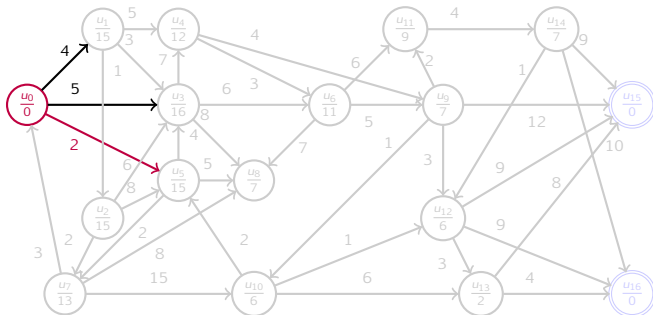
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$u(g(u), f(u))$	père(u)	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_1(4,19)u_3(5,21)$

$v = u_5$ $D = \{u_0\}$



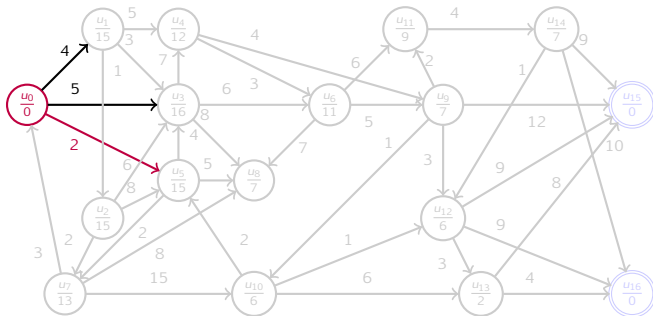
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$u(g(u), f(u))$	père(u)	$G = \{ v \mid v(g(v), f(v)) \}$
$u_0(0,0)$	$\perp$	$u_1(4,19)u_3(5,21)$

$v=u_5$ $D=\{u_0\}$



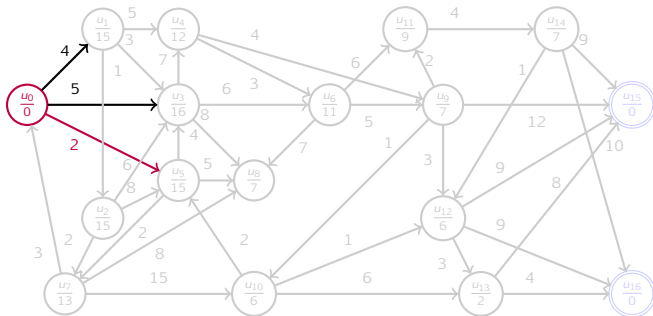
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$u_0(0,0)$	$\perp$	$u_1(4,19)u_3(5,21)$

$v=u_5$ $D=\{u_0\}$



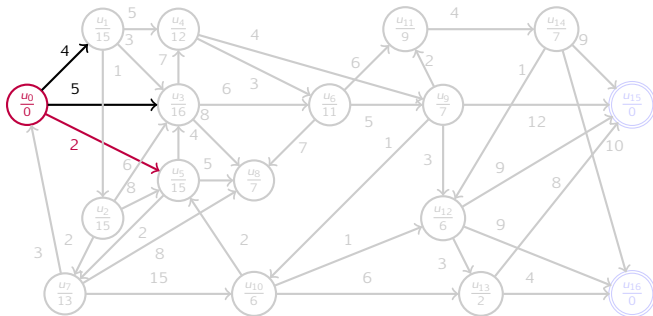
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$u(g(u), f(u))$	père(u)	$G = \{ v \mid v(g(v), f(v)) \}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$

$v = u_5$ $D = \{u_0\}$



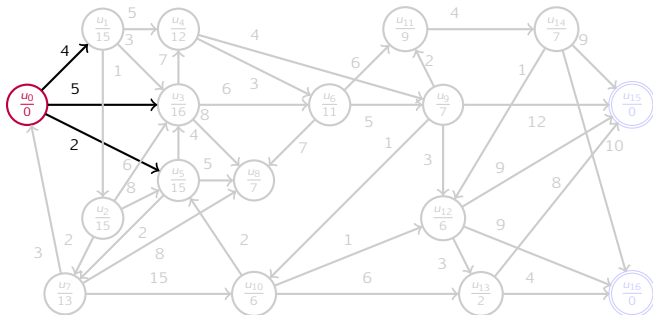
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$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$

$v = \perp$ D={u₀}



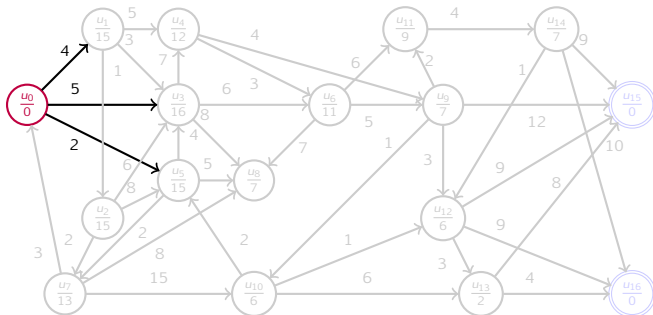
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$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$

$v = \perp$ D={ u_0 }



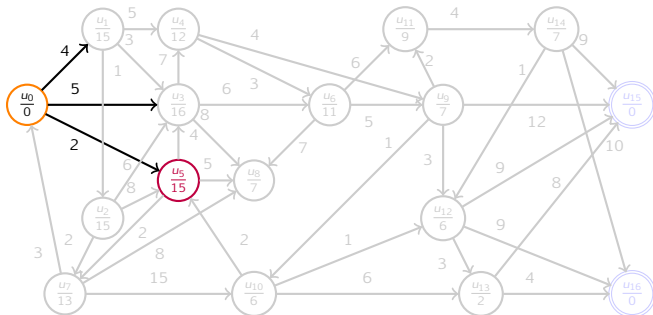
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$u_0(0,0)$ $u_5(2,17)$	$\perp$ u_0	$u_5(2,17)u_1(4,19)u_3(5,21)$

$v = \perp$ D={ u_0 }



Algorithme $A^*(u_0, \text{Sol})$

$$G \leftarrow \{u_0\}; D \leftarrow \emptyset; g(u_0) \leftarrow 0; f(u_0) \leftarrow 0; u \leftarrow \text{premier}(G);$$

```
while ( $G \neq \emptyset$ ) et ( $u \notin T$ ) do
```

$$G \leftarrow G \setminus \{u\}; D \leftarrow D \cup \{u\}; v \leftarrow \text{suivant}(u);$$

while (*v est valide*) **do**

if $((v \notin (D \cup G)) \text{ ou } (g(v) > g(u) + k(u,v)))$ then

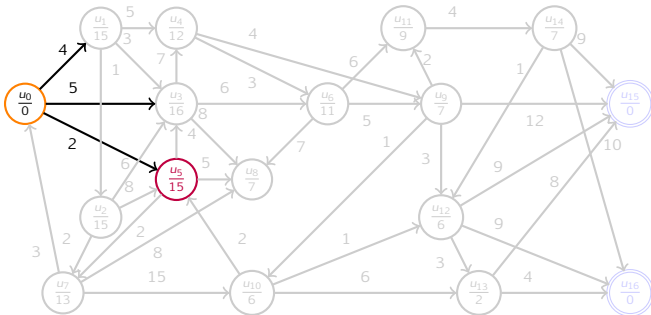
$$g(v) \leftarrow g(u) + k(u,v); f(v) \leftarrow g(v) + h(v);$$

$\text{père}(v) \leftarrow u : \text{insérer-selon-f}(G,v):$

```
v ← suivant(u);
```

```
u ← premier(G);
```

$u(g(u), f(u))$	$p\grave{e}re(u)$	$G = \{ v \mid v(g(v), f(v)) \}$
$u_0(0,0)$ $u_5(2,17)$	$\perp$ u_0	$u_5(2,17)u_1(4,19)u_3(5,21)$

$$v = \perp \quad D = \{u_0\}$$


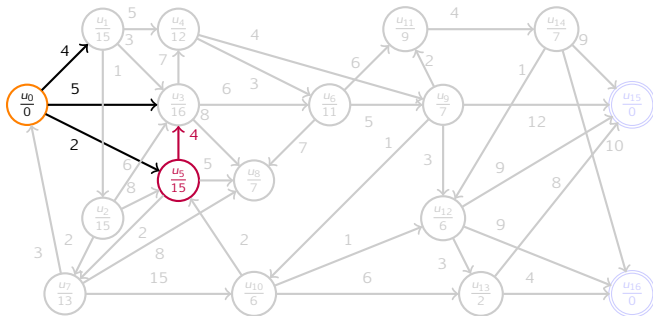
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$u(g(u), f(u))$	père(u)	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17) u_1(4,19) u_3(5,21)$
$u_5(2,17)$	u_0	$u_1(4,19) u_3(5,21)$

$v = u_3$ $D = \{u_0, u_5\}$



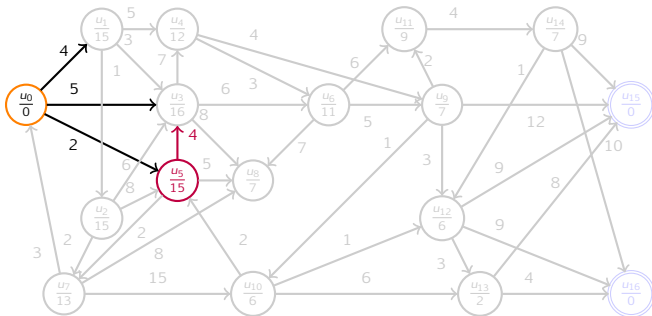
Algorithm A*(u₀, Sol)

```

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    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
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$u(g(u), f(u))$	père(u)	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17) u_1(4,19) u_3(5,21)$
$u_5(2,17)$	u_0	$u_1(4,19) u_3(5,21)$

$v = u_3$ $D = \{u_0, u_5\}$



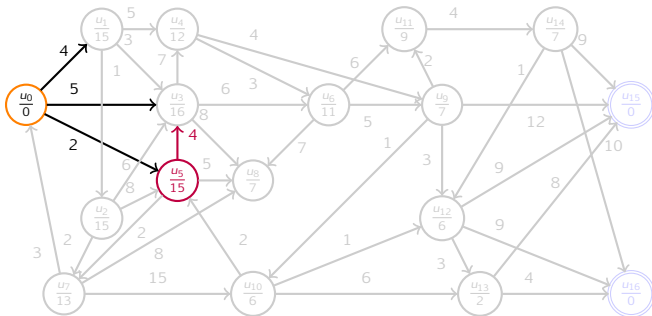
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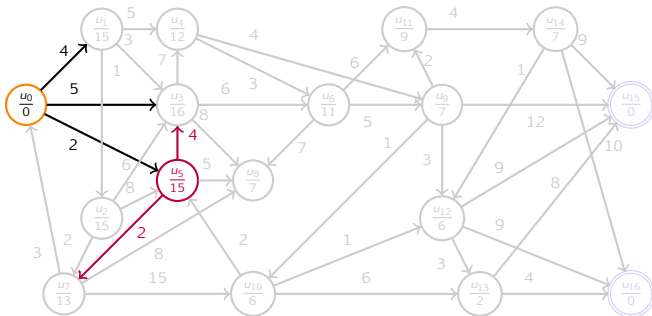
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$v = u_7$ $D = \{u_0, u_5\}$



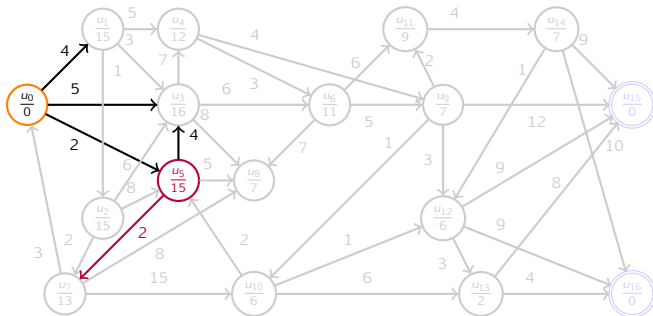
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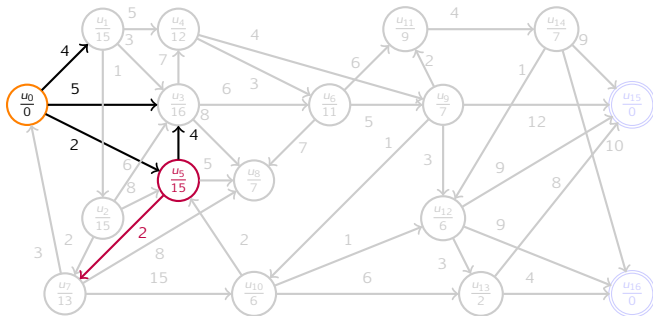
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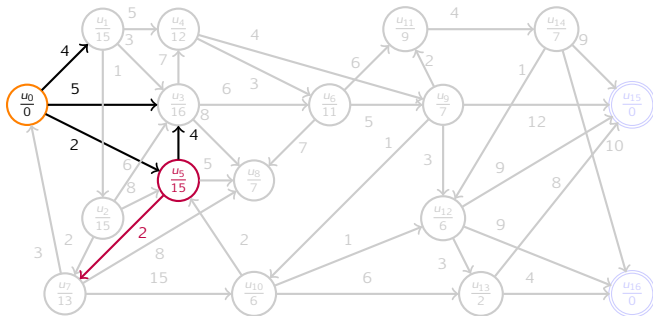
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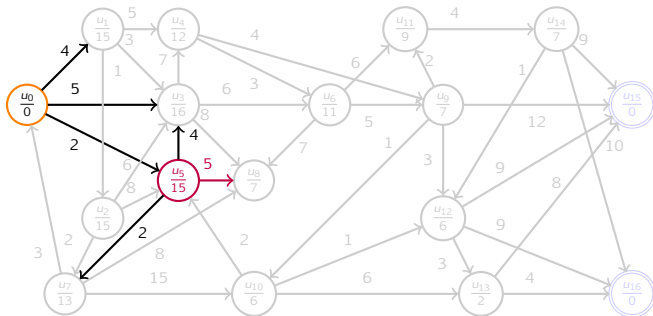
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$v = u_8$ $D = \{u_0, u_5\}$



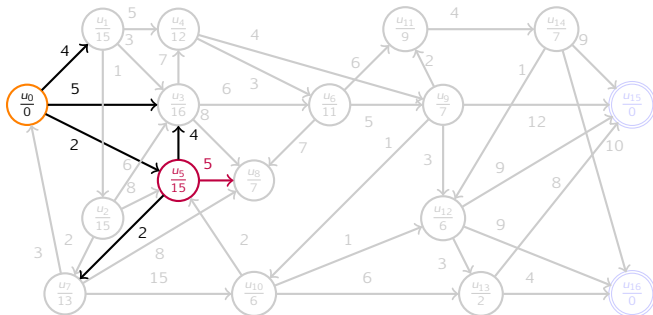
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$v=u_8$ $D=\{u_0, u_5\}$



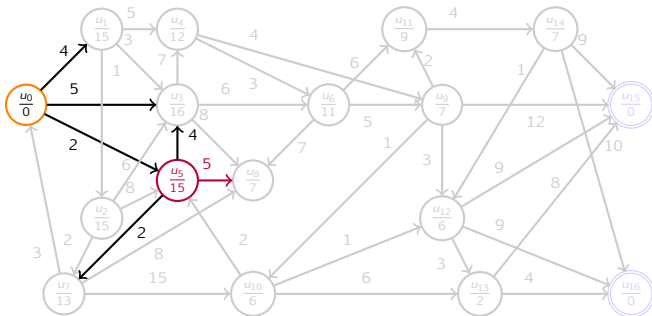
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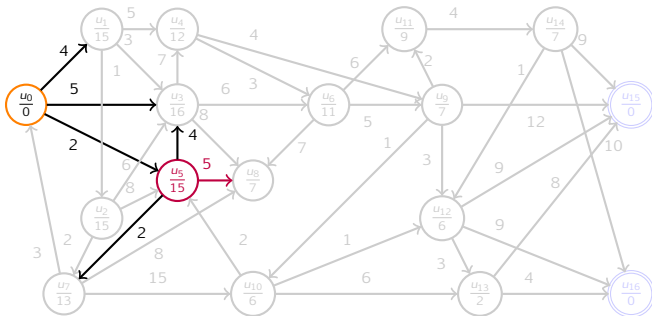
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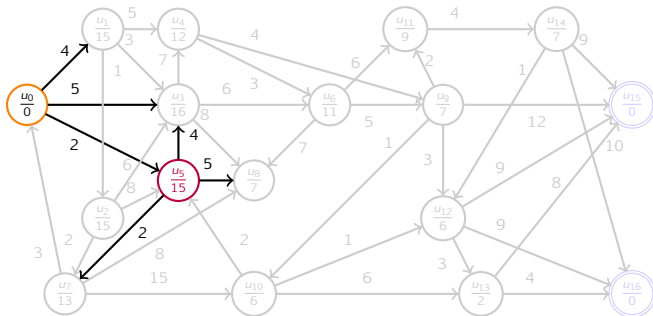
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$v = \perp$ D={u₀, u₅}



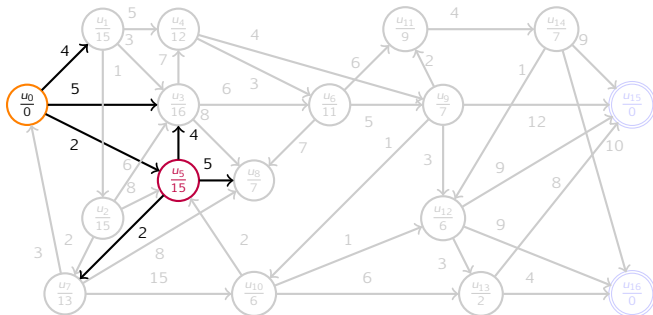
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$v = \perp$ D={u₀, u₅}



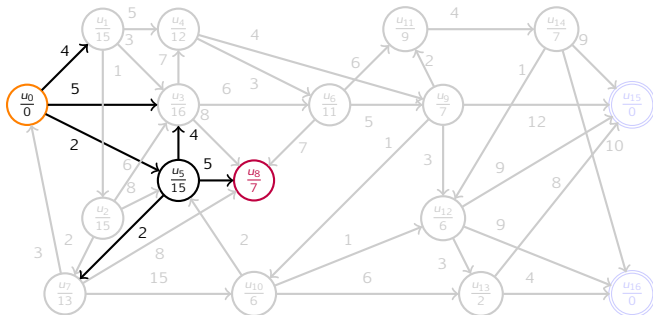
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$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	

$v = \perp$ D={u₀, u₅}



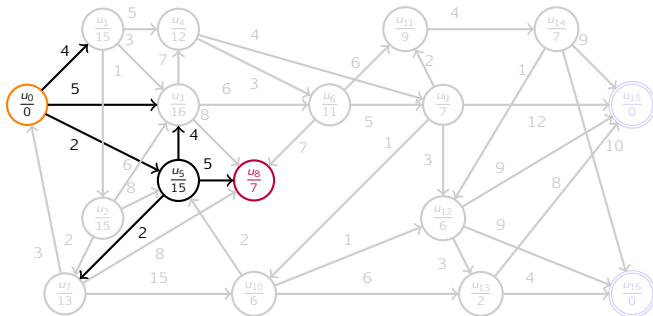
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$u_8(7,14)$	u_5	

$v = \perp$ D={u₀, u₅}



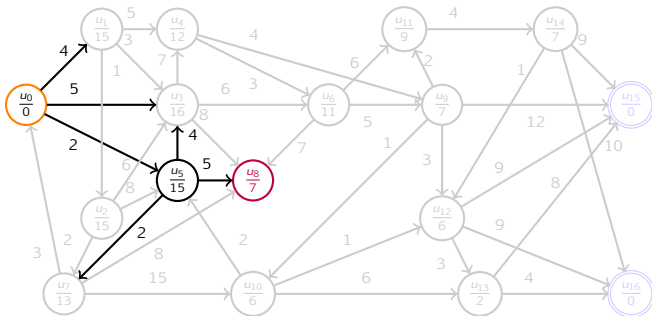
Algorithm A*(u_0, Sol)

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$v = \perp$ D={ u_0, u_5, u_8 }



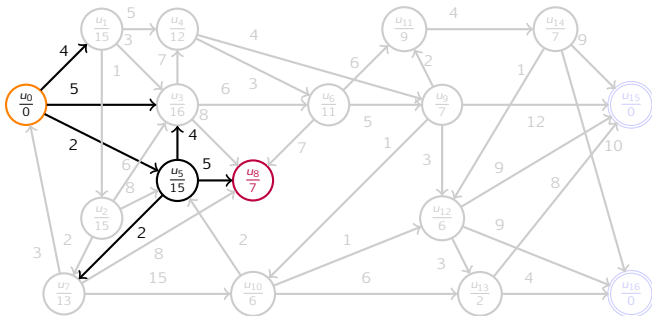
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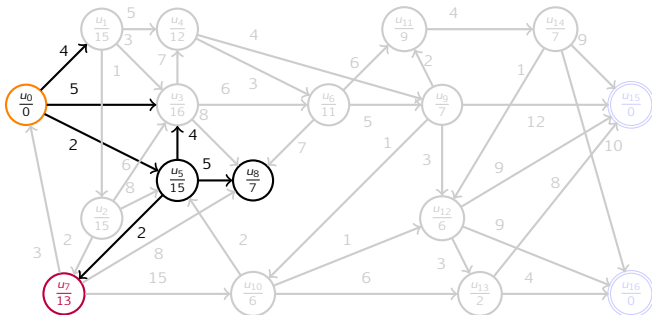
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$u_8(7,14)$	u_5	
$u_7(4,17)$	u_5	

$v = \perp$ D={u₀, u₅, u₈}



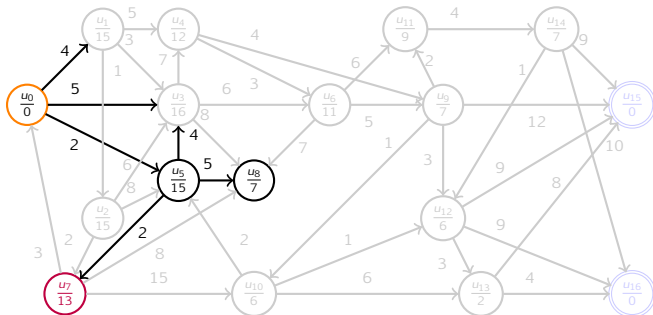
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```

$u(g(u), f(u))$	$père(u)$	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17) u_1(4,19) u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14) u_7(4,17) u_1(4,19) u_3(5,21)$
$u_8(7,14)$	u_5	
$u_7(4,17)$	u_5	

$v = \perp$ D={u₀, u₅, u₈}



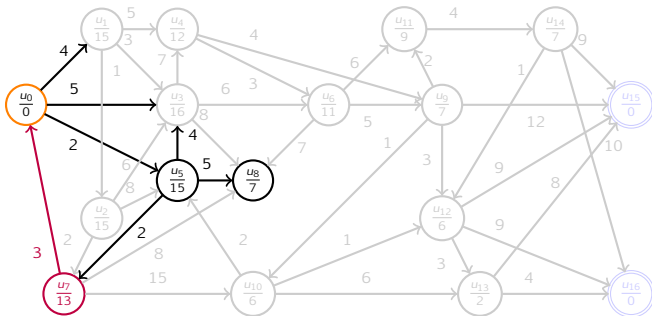
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅ et (u ∉ T)) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

$u(g(u), f(u))$	$père(u)$	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17) u_1(4,19) u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14) u_7(4,17) u_1(4,19) u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17) u_1(4,19) u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19) u_3(5,21)$

$v = u_0$ $D = \{u_0, u_5, u_7, u_8\}$



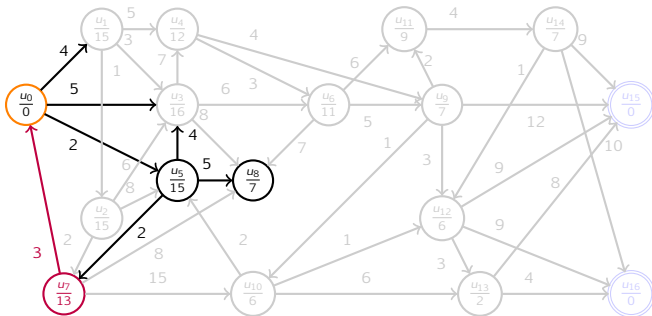
Algorithme A*(u₀,Sol)

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            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
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        v ← suivant(u);
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```

$u(g(u),f(u))$	père(u)	$G = \{v \mid v(g(v),f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)$

$v=u_0$ $D=\{u_0, u_5, u_7, u_8\}$



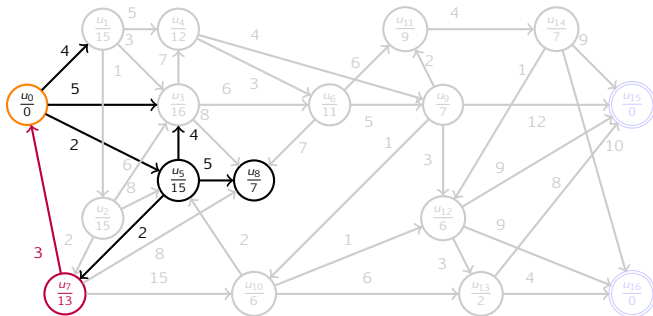
Algorithme $A^*(u_0, \text{Sol})$

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∈ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);

```

$u(g(u), f(u))$	père(u)	$G = \{ v \mid v(g(v), f(v)) \}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)$

$$v=u_0 \quad D=\{u_0, u_5, u_7, u_8\}$$


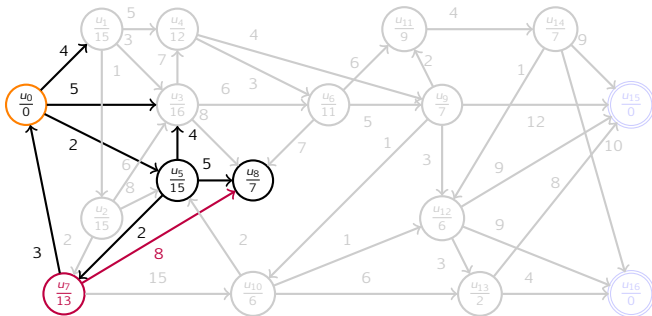
Algorithme A*(u₀, Sol)

```

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            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

$u(g(u), f(u))$	père(u)	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17) u_1(4,19) u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14) u_7(4,17) u_1(4,19) u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17) u_1(4,19) u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19) u_3(5,21)$

$v = u_8$ $D = \{u_0, u_5, u_7, u_8\}$



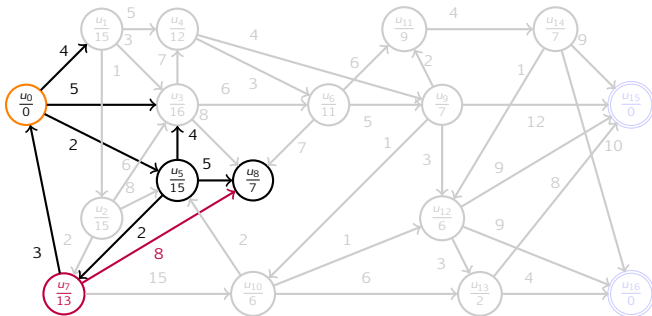
Algorithme A*(u₀,Sol)

```

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        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

$u(g(u),f(u))$	père(u)	$G = \{ v \mid v(g(v),f(v)) \}$
$u_0(0,0)$	$\perp$	$u_5(2,17) u_1(4,19) u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14) u_7(4,17) u_1(4,19) u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17) u_1(4,19) u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19) u_3(5,21)$

$v=u_8$ $D=\{u_0, u_5, u_7, u_8\}$



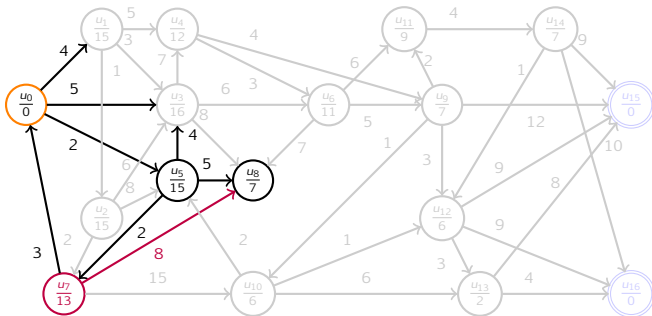
Algorithme $A^*(u_0, \text{Sol})$

```

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        v ← suivant(u);
    u ← premier(G);
    
```

$u(g(u), f(u))$	père(u)	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17) u_1(4,19) u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14) u_7(4,17) u_1(4,19) u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17) u_1(4,19) u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19) u_3(5,21)$

$v = u_8$ $D = \{u_0, u_5, u_7, u_8\}$



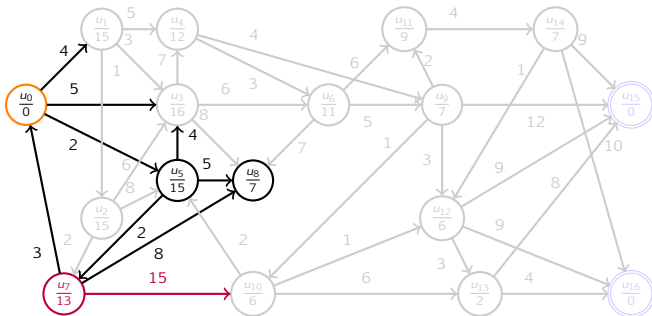
Algorithme $A^*(u_0, \text{Sol})$

```

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        v ← suivant(u);
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```

$u(g(u), f(u))$	père(u)	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17) u_1(4,19) u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14) u_7(4,17) u_1(4,19) u_3(5,21)$
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$u_7(4,17)$	u_5	$u_1(4,19) u_3(5,21)$

$v = u_{10}$ $D = \{u_0, u_5, u_7, u_8\}$



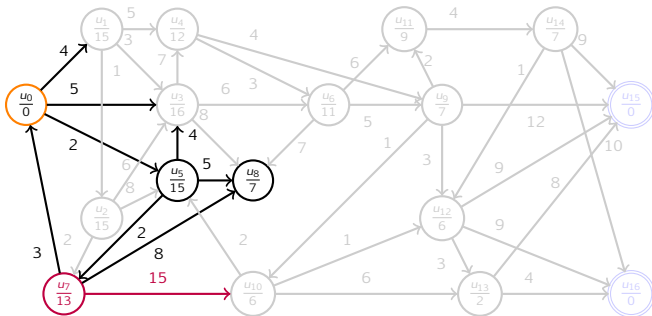
Algorithme $A^*(u_0, \text{Sol})$

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```

$u(g(u), f(u))$	père(u)	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17) u_1(4,19) u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14) u_7(4,17) u_1(4,19) u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17) u_1(4,19) u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19) u_3(5,21)$

$v = u_{10}$ $D = \{u_0, u_5, u_7, u_8\}$



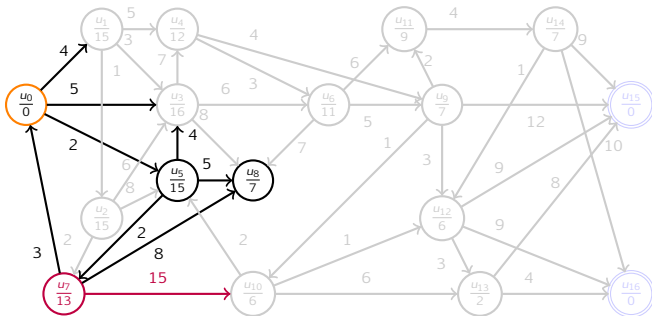
Algorithme $A^*(u_0, \text{Sol})$

```

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        v ← suivant(u);
    u ← premier(G);
    
```

$u(g(u), f(u))$	père(u)	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17) u_1(4,19) u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14) u_7(4,17) u_1(4,19) u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17) u_1(4,19) u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19) u_3(5,21)$

$v = u_{10}$ $D = \{u_0, u_5, u_7, u_8\}$



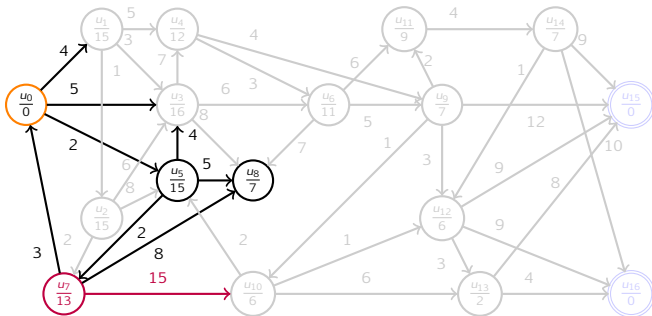
Algorithme A*(u₀,Sol)

```

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```

$u(g(u),f(u))$	père(u)	$G = \{v \mid v(g(v),f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$

$v=u_{10}$ D={u₀, u₅, u₇, u₈}



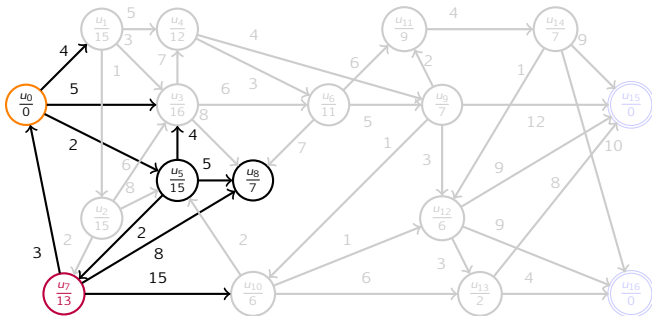
Algorithme $A^*(u_0, \text{Sol})$

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        v ← suivant(u);
    u ← premier(G);
    
```

$u(g(u), f(u))$	père(u)	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17) u_1(4,19) u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14) u_7(4,17) u_1(4,19) u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17) u_1(4,19) u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19) u_3(5,21) u_{10}(19,25)$

$v = \perp$ D={u₀, u₅, u₇, u₈}



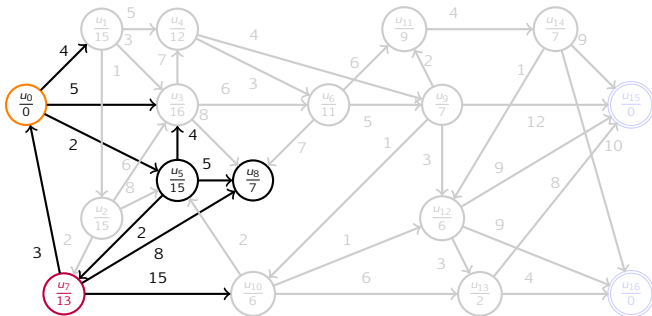
Algorithme $A^*(u_0, \text{Sol})$

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            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

$u(g(u), f(u))$	$\text{père}(u)$	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17) u_1(4,19) u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14) u_7(4,17) u_1(4,19) u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17) u_1(4,19) u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19) u_3(5,21) u_{10}(19,25)$

$v = \perp$ D={u₀, u₅, u₇, u₈}



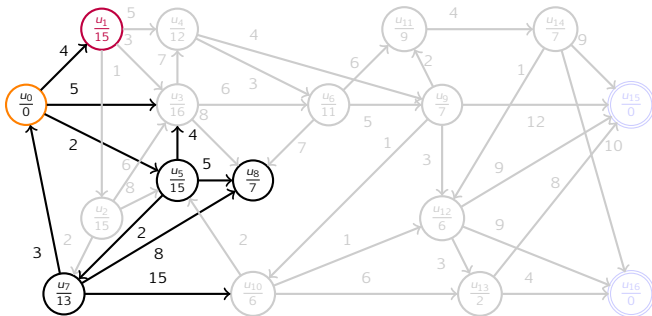
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅ et (u ∉ T)) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
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            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

v ← ∅ D ← {u₀, u₅, u₇, u₈}

u(g(u),f(u))	père(u)	G = { v v(g(v),f(v)) }
u ₀ (0,0)	⊥	u ₅ (2,17) u ₁ (4,19) u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14) u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21) u ₁₀ (19,25)
u ₁ (4,19)	u ₀	



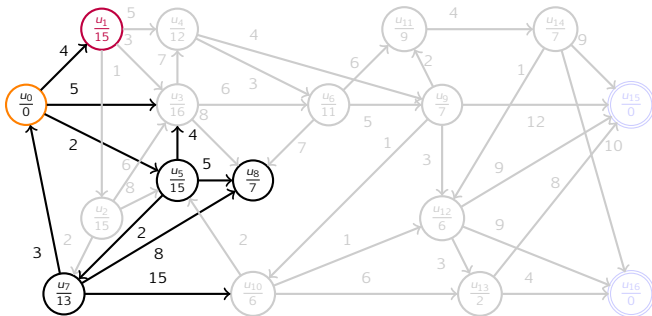
Algorithme A*(u₀, Sol)

```

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            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

$u(g(u), f(u))$	$père(u)$	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17) u_1(4,19) u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14) u_7(4,17) u_1(4,19) u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17) u_1(4,19) u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19) u_3(5,21) u_{10}(19,25)$
$u_1(4,19)$	u_0	

$v = \perp$ D={u₀, u₅, u₇, u₈}



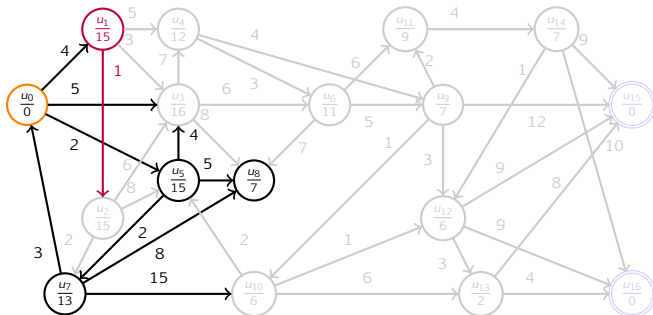
Algorithme $A^*(u_0, \text{Sol})$

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅ et (u ∉ T)) do
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        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

$u(g(u), f(u))$	père(u)	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17) u_1(4,19) u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14) u_7(4,17) u_1(4,19) u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17) u_1(4,19) u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19) u_3(5,21) u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_3(5,21) u_{10}(19,25)$

$v = u_2$ $D = \{u_0, u_1, u_5, u_7, u_8\}$



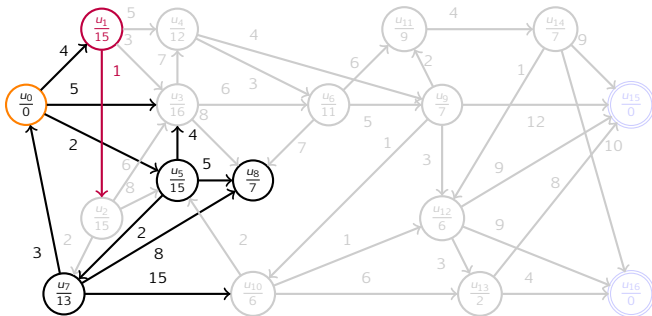
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
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```

$u(g(u), f(u))$	$père(u)$	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_3(5,21)u_{10}(19,25)$

$v = u_2$ $D = \{u_0, u_1, u_5, u_7, u_8\}$



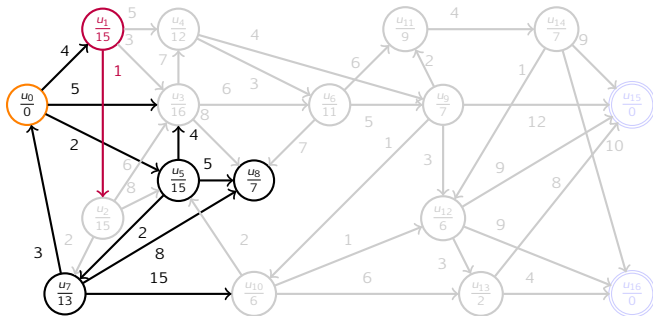
Algorithm A*(u₀, Sol)

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$u_5(2,17)$	u_0	$u_8(7,14) u_7(4,17) u_1(4,19) u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17) u_1(4,19) u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19) u_3(5,21) u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_3(5,21) u_{10}(19,25)$

$v = u_2$ $D = \{u_0, u_1, u_5, u_7, u_8\}$



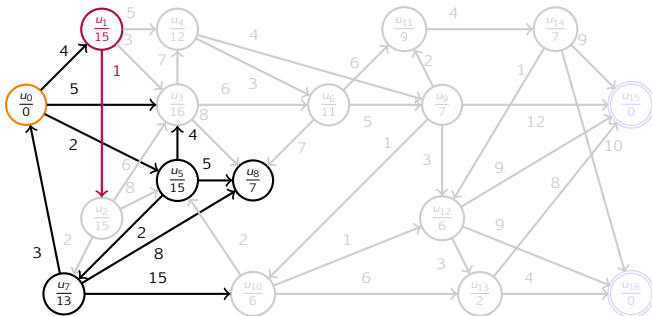
Algorithme A*(u₀,Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅ et (u ∉ T)) do
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```

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$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_3(5,21)u_{10}(19,25)$

$v=u_2$ $D=\{u_0, u_1, u_5, u_7, u_8\}$



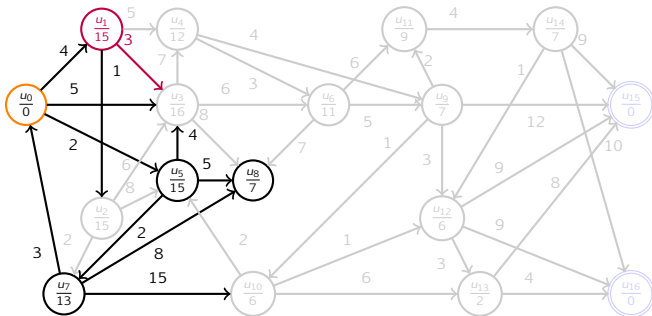
Algorithme $A^*(u_0, \text{Sol})$

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
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$u_5(2,17)$	u_0	$u_8(7,14) u_7(4,17) u_1(4,19) u_3(5,21)$
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$u_1(4,19)$	u_0	$u_2(5,20) u_3(5,21) u_{10}(19,25)$

$v = u_3$ $D = \{u_0, u_1, u_5, u_7, u_8\}$



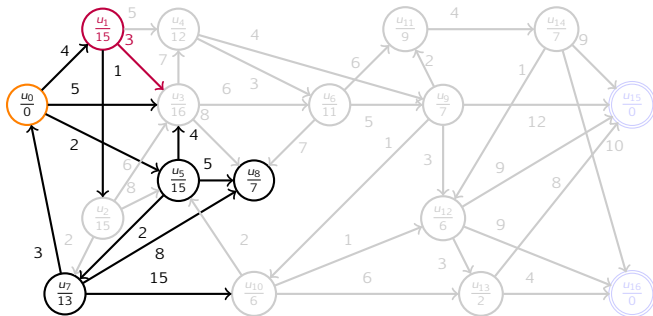
Algorithm A*(u₀, Sol)

```

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$u_8(7,14)$	u_5	$u_7(4,17) u_1(4,19) u_3(5,21)$
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$u_1(4,19)$	u_0	$u_2(5,20) u_3(5,21) u_{10}(19,25)$

$v = u_3$ $D = \{u_0, u_1, u_5, u_7, u_8\}$



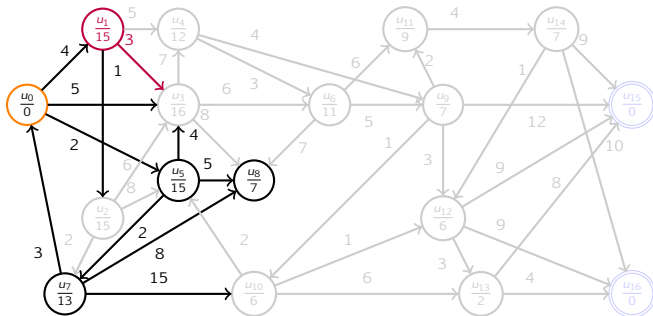
Algorithm A*(u₀, Sol)

```

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$v = u_3$ $D = \{u_0, u_1, u_5, u_7, u_8\}$



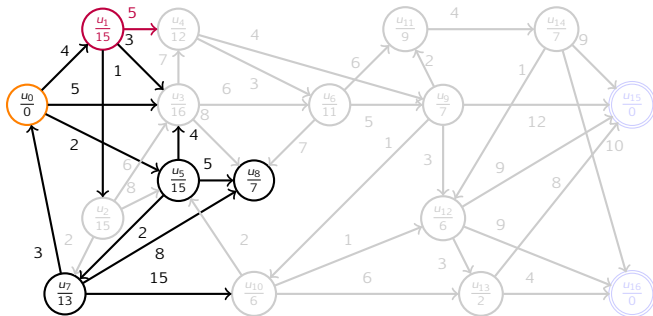
Algorithme $A^*(u_0, \text{Sol})$

```

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$u_1(4,19)$	u_0	$u_2(5,20) u_3(5,21) u_{10}(19,25)$

$v = u_4$ $D = \{u_0, u_1, u_5, u_7, u_8\}$



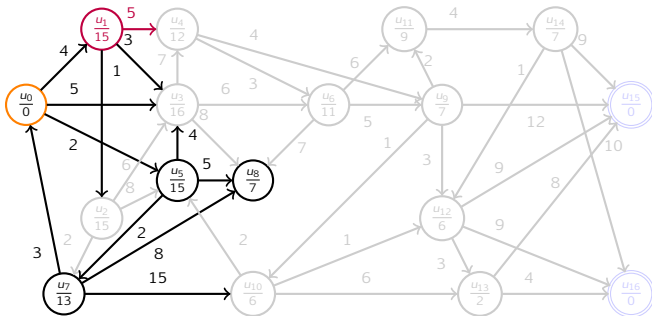
Algorithme $A^*(u_0, \text{Sol})$

```

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$v = u_4$ $D = \{u_0, u_1, u_5, u_7, u_8\}$



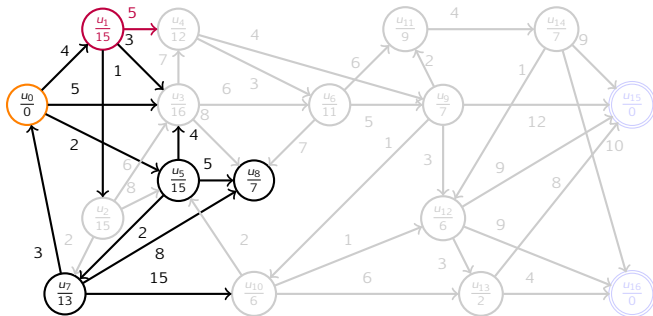
Algorithm A*(u_0, Sol)

```

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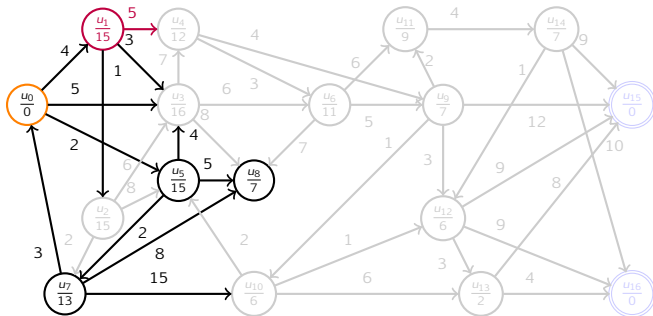
Algorithm A*(u₀, Sol)

```

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$v = u_4$ $D = \{u_0, u_1, u_5, u_7, u_8\}$



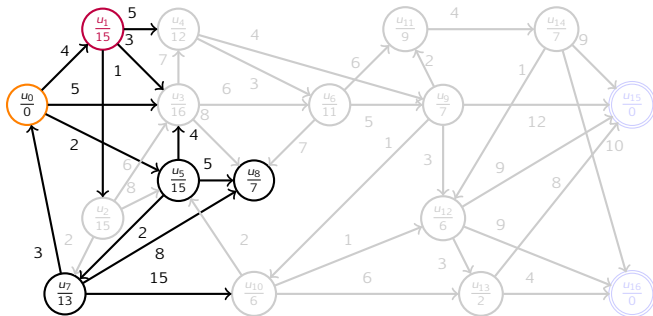
Algorithme $A^*(u_0, \text{Sol})$

```

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$v = \perp$ D={u₀, u₁, u₅, u₇, u₈}



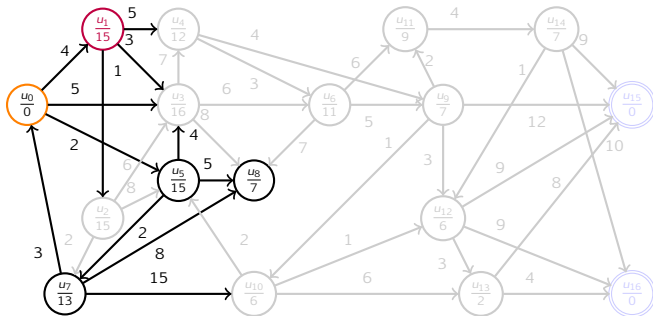
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```

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$v = \perp$ D={u₀, u₁, u₅, u₇, u₈}



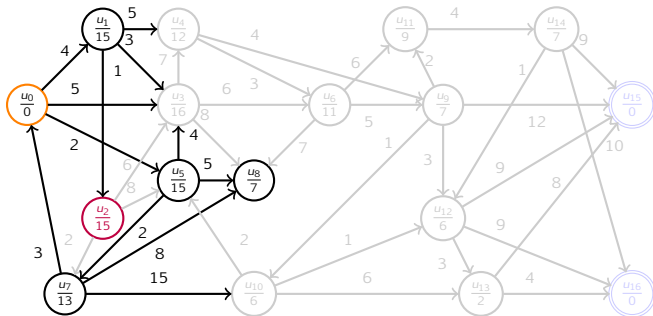
Algorithm A*(u₀, Sol)

```

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$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	

$v = \perp$ D={u₀, u₁, u₅, u₇, u₈}



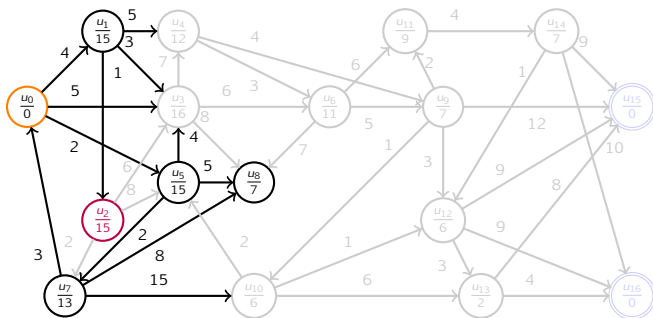
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅ et (u ∉ T)) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

$u(g(u), f(u))$	$père(u)$	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	

$v = \perp$ D={u₀, u₁, u₅, u₇, u₈}



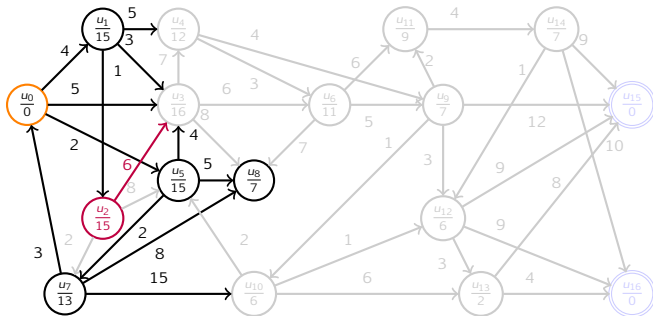
Algorithme $A^*(u_0, \text{Sol})$

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∈ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);

```

$u(g(u), f(u))$	$père(u)$	$G = \{ v \mid v(g(v), f(v)) \}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$

$$v = u_3 \quad D = \{u_0, u_1, u_2, u_5, u_7, u_8\}$$


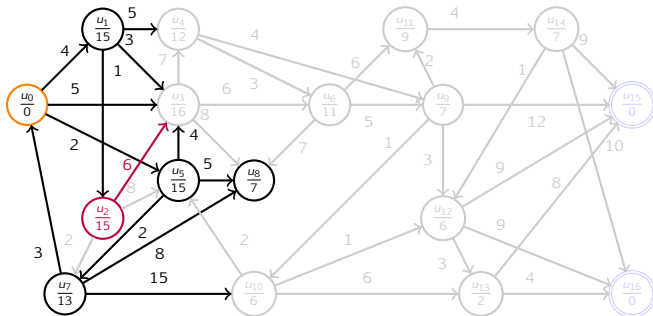
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅ et (u ∉ T)) do
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        v ← suivant(u);
    u ← premier(G);
    
```

$u(g(u), f(u))$	père(u)	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17) u_1(4,19) u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14) u_7(4,17) u_1(4,19) u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17) u_1(4,19) u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19) u_3(5,21) u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20) u_4(9,21) u_3(5,21) u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21) u_3(5,21) u_{10}(19,25)$

$v = u_3$ $D = \{u_0, u_1, u_2, u_5, u_7, u_8\}$



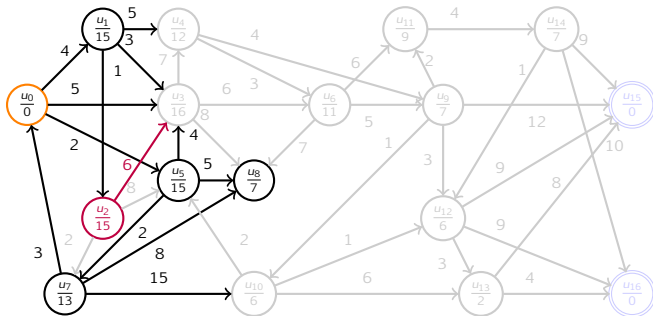
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅ et (u ∉ T)) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
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            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

$u(g(u), f(u))$	père(u)	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$

$v = u_3$ $D = \{u_0, u_1, u_2, u_5, u_7, u_8\}$



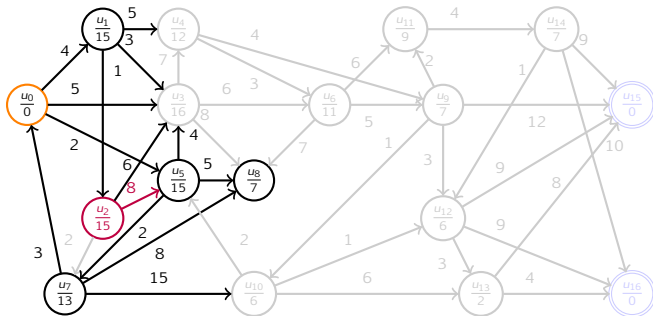
Algorithme A*(u₀,Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅ et (u ∉ T)) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
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            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
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        v ← suivant(u);
    u ← premier(G);
    
```

$u(g(u), f(u))$	père(u)	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17) u_1(4,19) u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14) u_7(4,17) u_1(4,19) u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17) u_1(4,19) u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19) u_3(5,21) u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20) u_4(9,21) u_3(5,21) u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21) u_3(5,21) u_{10}(19,25)$

$v = u_5$ $D = \{u_0, u_1, u_2, u_5, u_7, u_8\}$



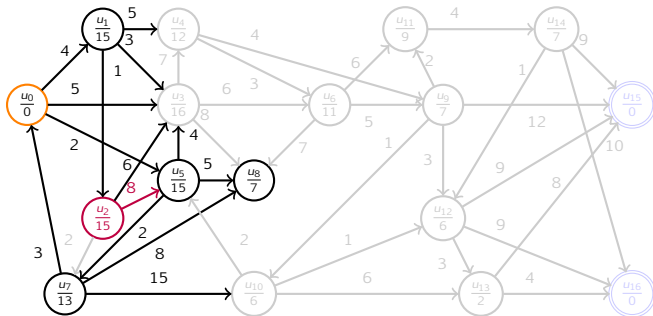
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
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            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

$u(g(u), f(u))$	père(u)	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
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$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$

$v = u_5$ $D = \{u_0, u_1, u_2, u_5, u_7, u_8\}$



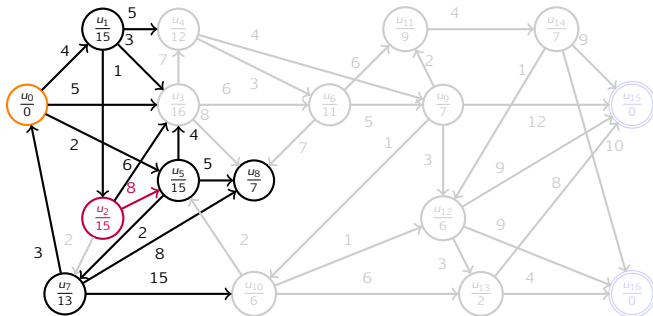
Algorithm A*(u₀, Sol)

```

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    u ← premier(G);
    
```

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$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$

$v = u_5$ $D = \{u_0, u_1, u_2, u_5, u_7, u_8\}$



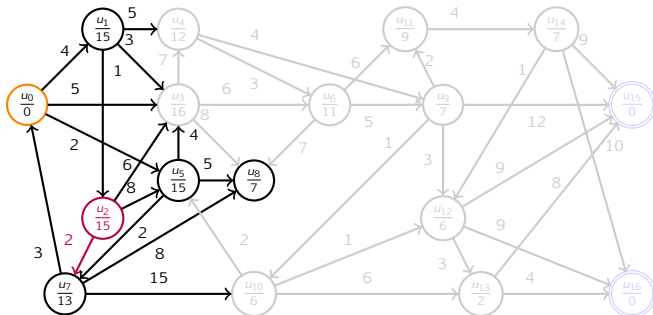
Algorithm A*(u₀, Sol)

```

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```

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$u_5(2,17)$	u_0	$u_8(7,14) u_7(4,17) u_1(4,19) u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17) u_1(4,19) u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19) u_3(5,21) u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20) u_4(9,21) u_3(5,21) u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21) u_3(5,21) u_{10}(19,25)$

$v = u_7$ $D = \{u_0, u_1, u_2, u_5, u_7, u_8\}$



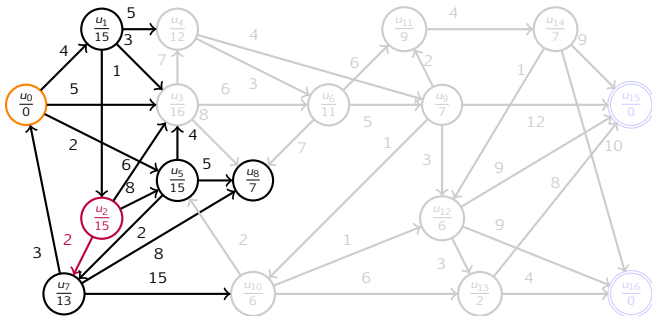
Algorithm A*(u₀, Sol)

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$u_2(5,20)$	u_1	$u_4(9,21) u_3(5,21) u_{10}(19,25)$

$v = u_7$ $D = \{u_0, u_1, u_2, u_5, u_7, u_8\}$



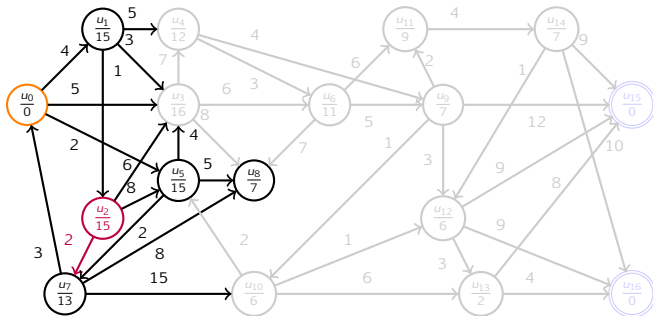
Algorithm A*(u₀, Sol)

```

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$v = u_7$ $D = \{u_0, u_1, u_2, u_5, u_7, u_8\}$



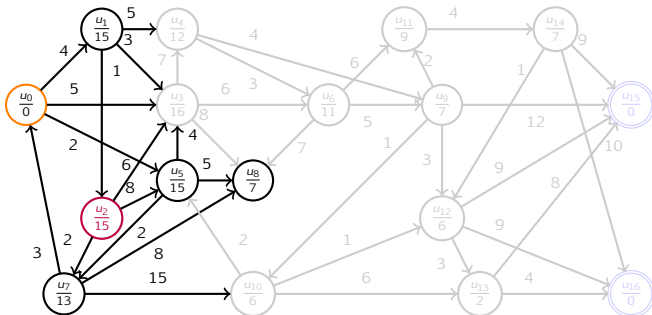
Algorithm A*(u₀, Sol)

```

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$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$

$v = \perp$ D={u₀, u₁, u₂, u₅, u₇, u₈}



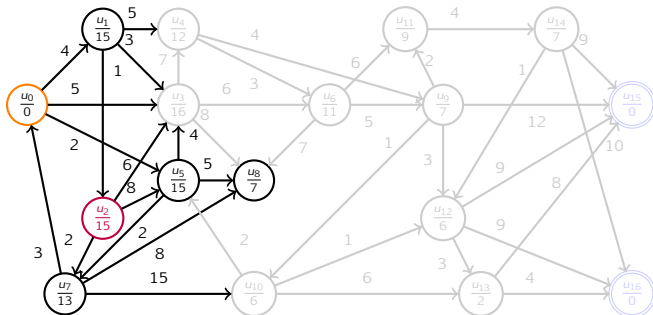
Algorithm A*(u₀, Sol)

```

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```

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$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$

$v = \perp$ D={u₀, u₁, u₂, u₅, u₇, u₈}



$u(g(u), f(u))$	$p\grave{e}re(u)$	$G = \{ v \mid v(g(v), f(v)) \}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
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$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$

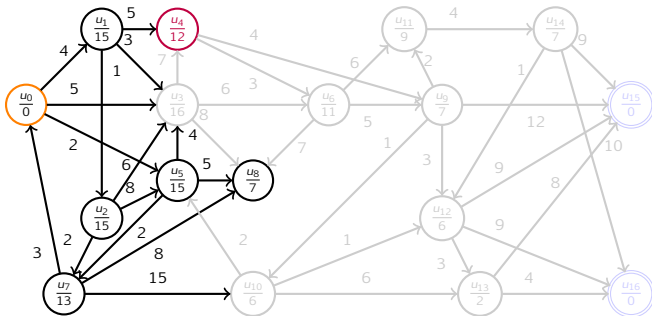
Algorithme A*(u₀,Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
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        v ← suivant(u);
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$u(g(u),f(u))$	$père(u)$	$G = \{v \mid v(g(v),f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_4(9,21)$	u_1	

$v = \perp$ D={u₀, u₁, u₂, u₅, u₇, u₈}



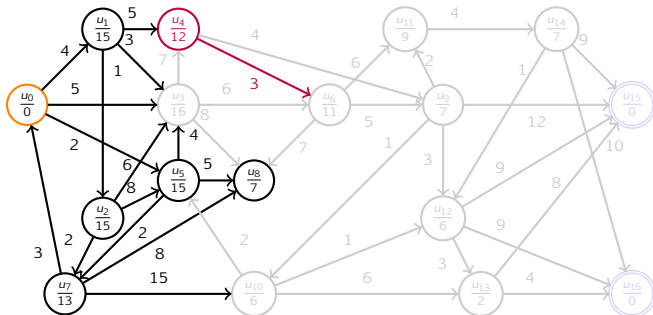
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$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_4(9,21)$	u_1	

$v = u_6$ $D = \{u_0, u_1, u_2, u_4, u_5, u_7, u_8\}$



$u(g(u), f(u))$	père(u)	$G = \{ v \mid v(g(v), f(v)) \}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
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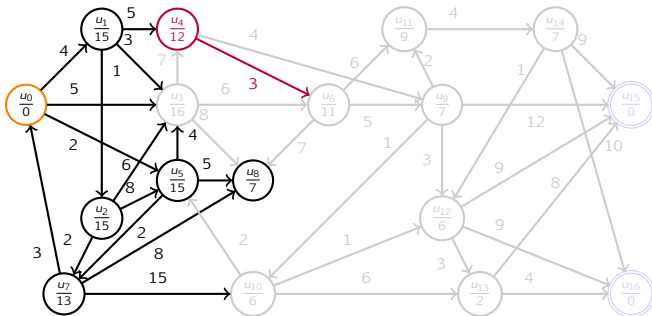
Algorithm A*(u₀, Sol)

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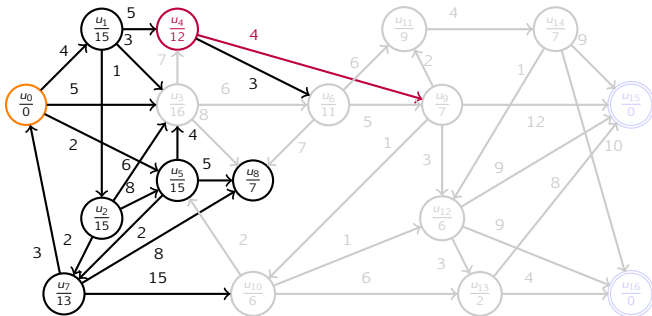
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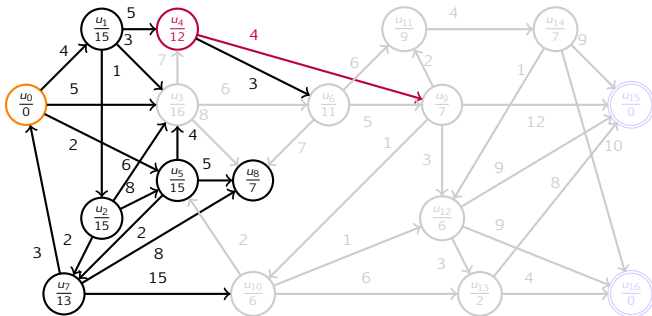
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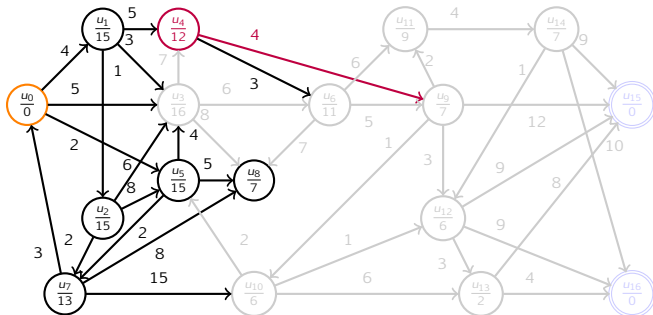
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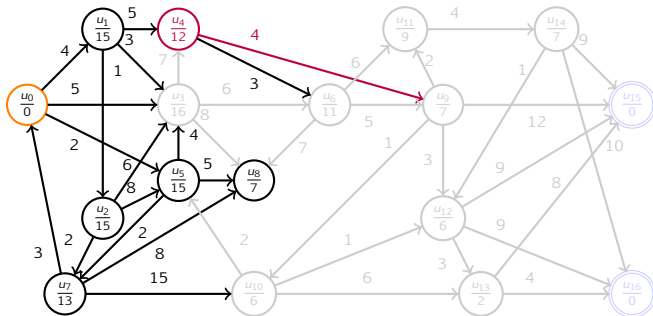
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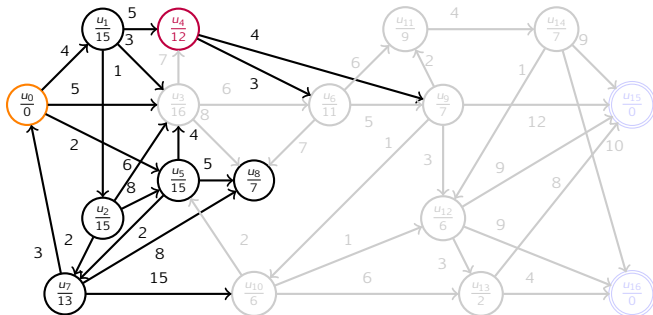
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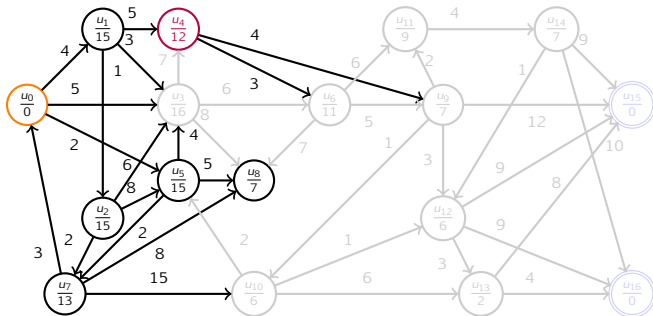
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$u_9(13,20)$	u_4	

The graph consists of 17 nodes, each labeled with a fraction u_i/d_i . The nodes are arranged in a roughly circular pattern. The nodes are: $u_0/0$ (orange border), $u_1/15$, $u_2/15$, $u_3/16$, $u_4/12$, $u_5/15$, $u_6/11$, $u_7/13$ (red border), $u_8/7$, $u_9/9$, $u_{10}/6$, $u_{11}/9$, $u_{12}/6$, $u_{13}/2$, $u_{14}/7$, $u_{15}/0$ (blue border), and $u_{16}/0$ (blue border). The edges are directed and labeled with weights. The graph shows a complex network of connections with various weights on the edges.

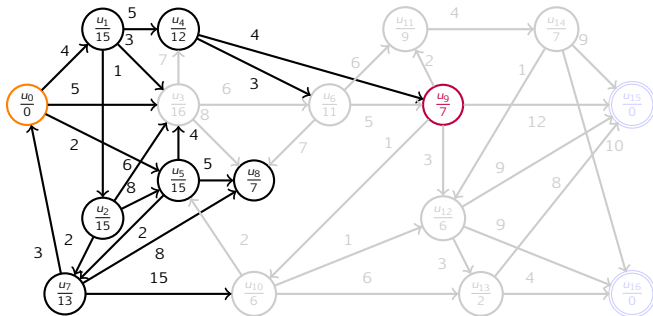
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$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20)u_3(5,21)u_6(12,23)u_{10}(19,25)$
$u_9(13,20)$	u_4	

$v = \perp$ D={u₀, u₁, u₂, u₄, u₅, u₇, u₈}



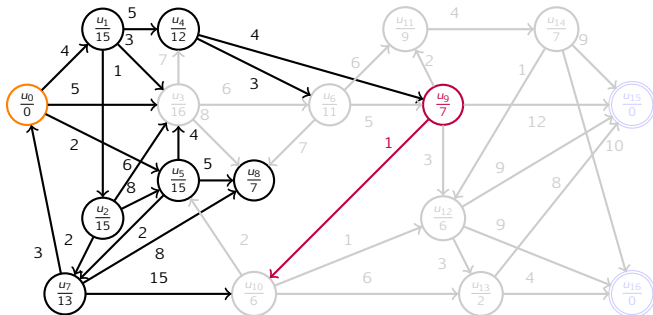
Algorithme A*(u₀,Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅ et (u ∉ T)) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

$u(g(u),f(u))$	père(u)	$G = \{v \mid v(g(v),f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20)u_3(5,21)u_6(12,23)u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_3(5,21)u_6(12,23)u_{10}(19,25)$

$v=u_{10}$ D={u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉}



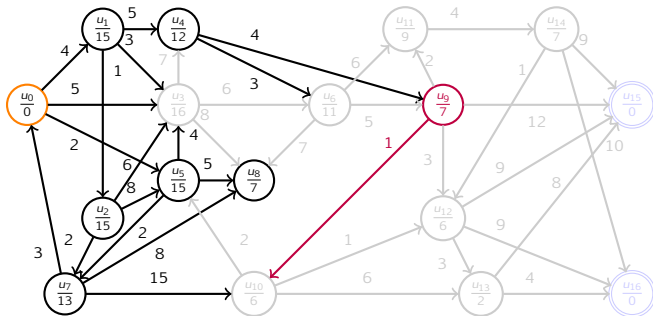
Algorithm A*(u₀, Sol)

```

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```

$u(g(u), f(u))$	père(u)	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
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$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$
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$v=u_{10}$ $D=\{u_0, u_1, u_2, u_4, u_5, u_7, u_8, u_9\}$



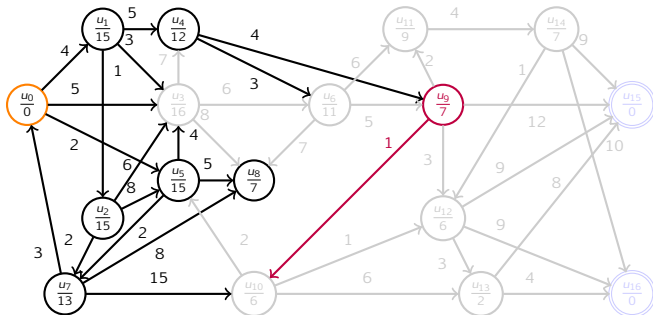
Algorithm A*(u₀, Sol)

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$u(g(u), f(u))$	père(u)	$G = \{v \mid v(g(v), f(v))\}$
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$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20)u_3(5,21)u_6(12,23)u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_3(5,21)u_6(12,23)u_{10}(19,25)$

$v=u_{10}$ $D=\{u_0, u_1, u_2, u_4, u_5, u_7, u_8, u_9\}$



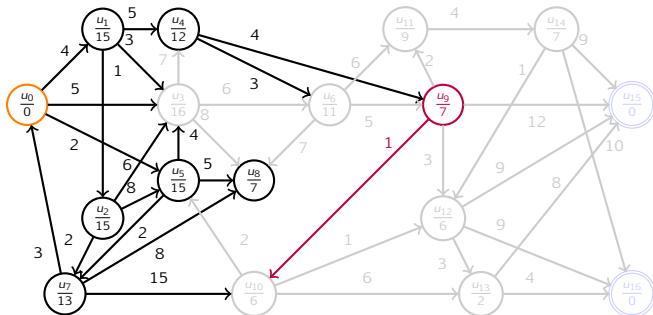
Algorithm A*(u₀, Sol)

```

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$u(g(u), f(u))$	$père(u)$	$G = \{v \mid v(g(v), f(v))\}$
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$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$
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$v=u_{10}$ $D=\{u_0, u_1, u_2, u_4, u_5, u_7, u_8, u_9\}$



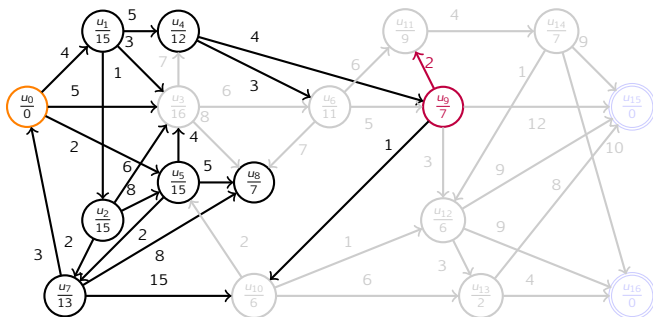
Algorithm A*(u₀, Sol)

```

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$u_9(13,20)$	u_4	$u_{10}(14,20)u_3(5,21)u_6(12,23)$

$v = u_{11}$ $D = \{u_0, u_1, u_2, u_4, u_5, u_7, u_8, u_9\}$



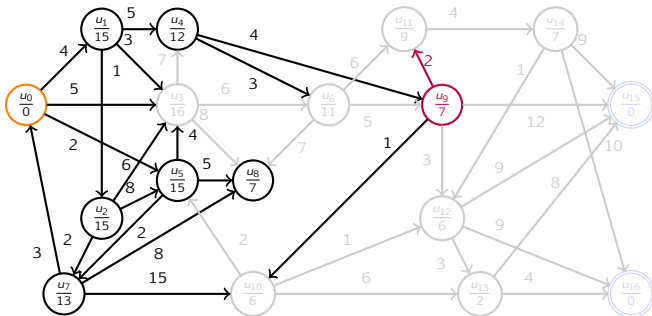
Algorithm A*(u₀, Sol)

```

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$v=u_{11}$ $D=\{u_0, u_1, u_2, u_4, u_5, u_7, u_8, u_9\}$



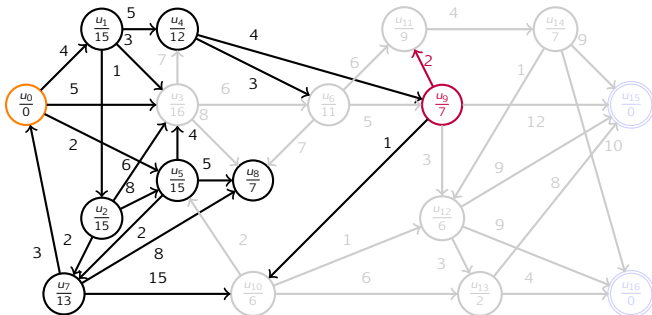
Algorithm A*(u₀, Sol)

```

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$u(g(u), f(u))$	père(u)	$G = \{v \mid v(g(v), f(v))\}$
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$v=u_{11}$ $D=\{u_0, u_1, u_2, u_4, u_5, u_7, u_8, u_9\}$



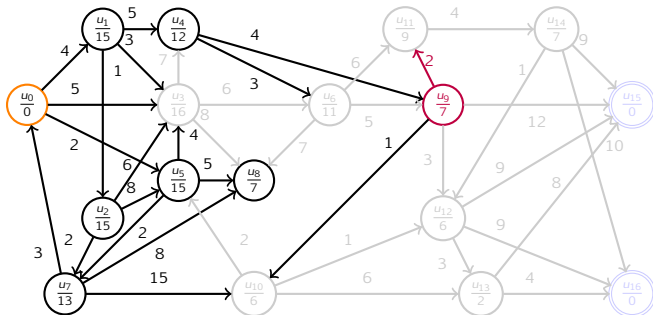
Algorithm A*(u₀, Sol)

```

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$u_9(13,20)$	u_4	$u_{10}(14,20)u_3(5,21)u_6(12,23)u_{11}(15,24)$

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$u(g(u), f(u))$	$p\grave{e}re(u)$	$G = \{ v \mid v(g(v), f(v)) \}$
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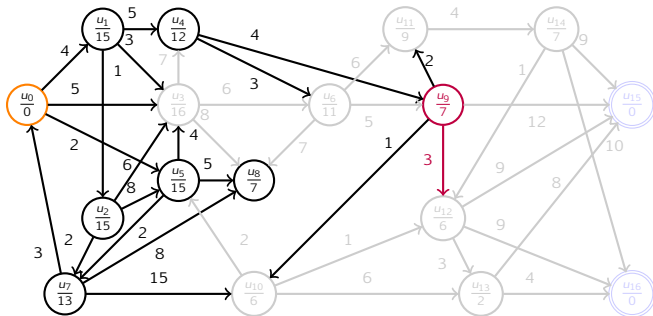
Algorithm A*(u₀, Sol)

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$v=u_{12}$ D={u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉}



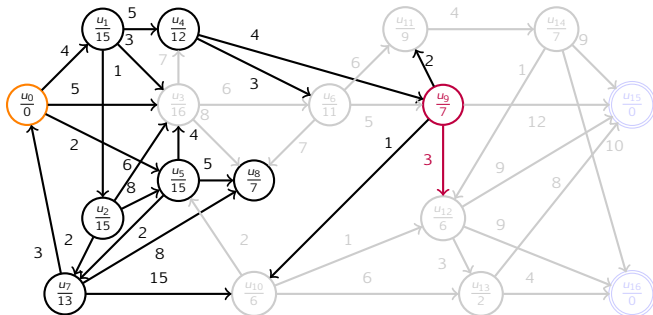
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$v=u_{12}$ $D=\{u_0, u_1, u_2, u_4, u_5, u_7, u_8, u_9\}$



Algorithm A*(u₀, Sol)

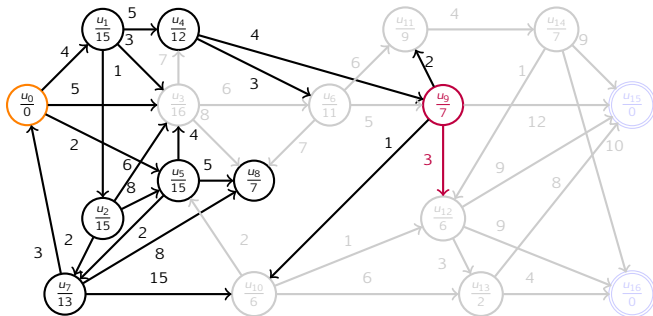
```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅ et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);

```

v = u₁₂ D = {u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉}

u(g(u), f(u))	père(u)	G = { v v(g(v), f(v)) }
u ₀ (0,0)	⊥	u ₅ (2,17) u ₁ (4,19) u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14) u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21) u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20) u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20) u ₃ (5,21) u ₆ (12,23) u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20) u ₃ (5,21) u₁₂(16,22) u ₆ (12,23)
		u ₁₁ (15,24)



Algorithm A*(u₀, Sol)

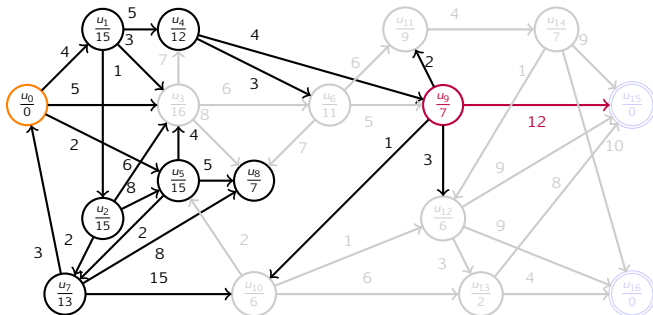
```

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        v ← suivant(u);
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```

v = u₁₅ D = {u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉}

u(g(u),f(u))	père(u)	G = { v v(g(v),f(v)) }
u ₀ (0,0)	⊥	u ₅ (2,17) u ₁ (4,19) u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14) u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21) u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20) u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20) u ₃ (5,21) u ₆ (12,23) u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20) u ₃ (5,21) u ₁₂ (16,22) u ₆ (12,23)
		u ₁₁ (15,24)



Algorithm A*(u₀, Sol)

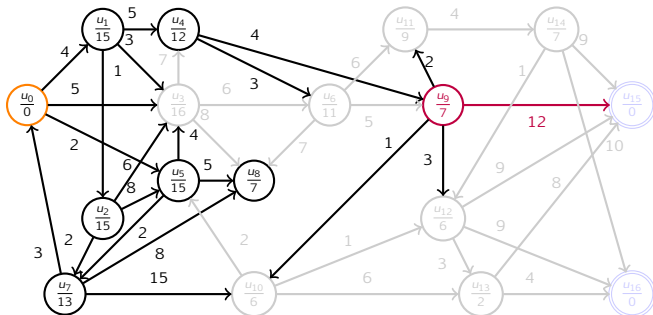
```

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            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);

```

v = u₁₅ D = {u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉}

u(g(u), f(u))	père(u)	G = { v v(g(v), f(v)) }
u ₀ (0,0)	⊥	u ₅ (2,17) u ₁ (4,19) u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14) u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21) u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20) u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20) u ₃ (5,21) u ₆ (12,23) u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20) u ₃ (5,21) u ₁₂ (16,22) u ₆ (12,23)
		u ₁₁ (15,24)



Algorithm A*(u₀, Sol)

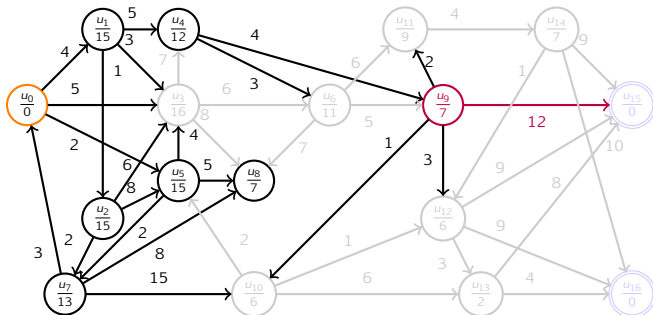
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            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
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```

v = u₁₅ D = {u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉}

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u ₅ (2,17)	u ₀	u ₈ (7,14) u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21) u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20) u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20) u ₃ (5,21) u ₆ (12,23) u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20) u ₃ (5,21) u ₁₂ (16,22) u ₆ (12,23)
		u ₁₁ (15,24)



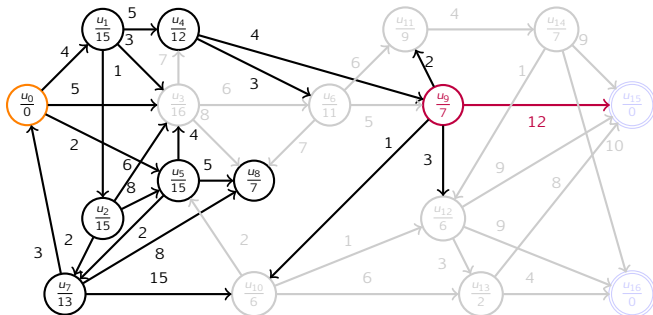
Algorithm A*(u₀, Sol)

```

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        v ← suivant(u);
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```

v=u₁₅ D={u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉}

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u ₀ (0,0)	⊥	u ₅ (2,17)u ₁ (4,19)u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14)u ₇ (4,17)u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21)u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20)u ₄ (9,21)u ₃ (5,21)u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21)u ₃ (5,21)u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20)u ₃ (5,21)u ₆ (12,23)u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20)u ₃ (5,21)u ₁₂ (16,22) u ₆ (12,23)
		u ₁₁ (15,24) u₁₅(25,25)



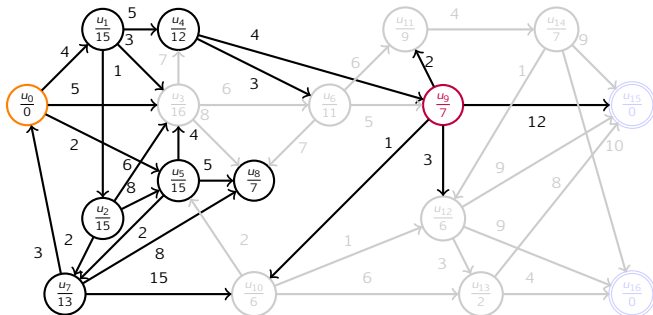
Algorithm A*(u₀, Sol)

```

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```

v ← $\perp$ D ← {u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉}

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u ₀ (0,0)	$\perp$	u ₅ (2,17) u ₁ (4,19) u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14) u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21) u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20) u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20) u ₃ (5,21) u ₆ (12,23) u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20) u ₃ (5,21) u ₁₂ (16,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)



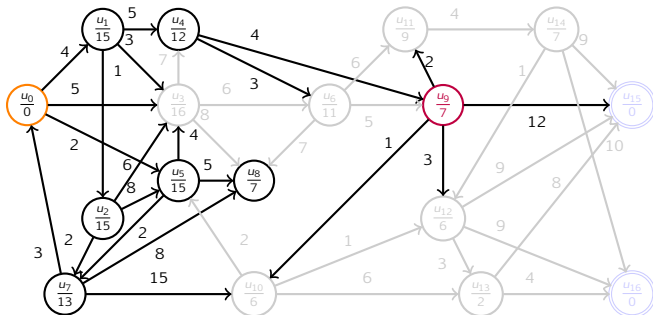
Algorithm A*(u₀, Sol)

```

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$u(g(u), f(u))$	$père(u)$	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17) u_1(4,19) u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14) u_7(4,17) u_1(4,19) u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17) u_1(4,19) u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19) u_3(5,21) u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20) u_4(9,21) u_3(5,21) u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21) u_3(5,21) u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20) u_3(5,21) u_6(12,23) u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20) u_3(5,21) u_{12}(16,22) u_6(12,23)$
		$u_{11}(15,24) u_{15}(25,25)$



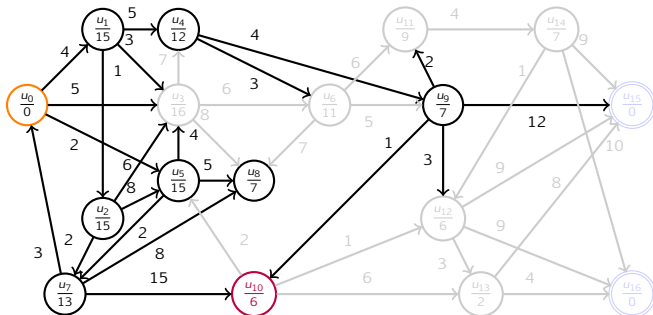
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```

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$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20)u_3(5,21)u_6(12,23)u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20)u_3(5,21)u_{12}(16,22)u_6(12,23)$
$u_{10}(14,20)$	u_9	$u_{11}(15,24)u_{15}(25,25)$



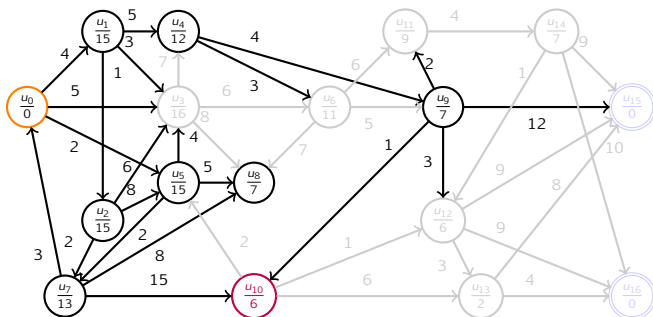
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$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20)u_3(5,21)u_6(12,23)u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20)u_3(5,21)u_{12}(16,22)u_6(12,23)$
$u_{10}(14,20)$	u_9	$u_{11}(15,24)u_{15}(25,25)$



Algorithm A*(u₀, Sol)

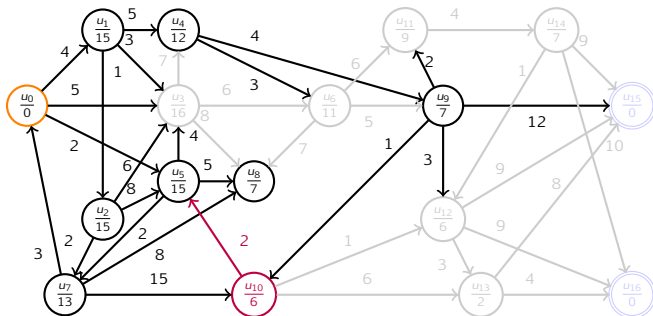
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        v ← suivant(u);
    u ← premier(G);

```

v = u₅ D = {u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀}

$u(g(u), f(u))$	père(u)	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17) u_1(4,19) u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14) u_7(4,17) u_1(4,19) u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17) u_1(4,19) u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19) u_3(5,21) u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20) u_4(9,21) u_3(5,21) u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21) u_3(5,21) u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20) u_3(5,21) u_6(12,23) u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20) u_3(5,21) u_{12}(16,22) u_6(12,23)$
$u_{10}(14,20)$	u_9	$u_{11}(15,24) u_{15}(25,25)$
		$u_3(5,21) u_{12}(16,22) u_6(12,23) u_{11}(15,24)$
		$u_{15}(25,25)$



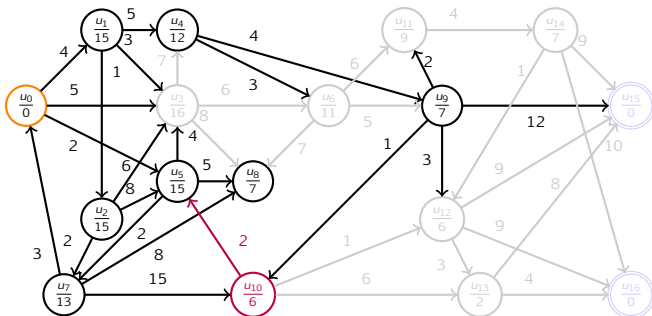
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

v = u₅ D = {u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀}

u(g(u), f(u))	père(u)	G = { v v(g(v), f(v)) }
u ₀ (0,0)	⊥	u ₅ (2,17) u ₁ (4,19) u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14) u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21) u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20) u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20) u ₃ (5,21) u ₆ (12,23) u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20) u ₃ (5,21) u ₁₂ (16,22) u ₆ (12,23)
u ₁₀ (14,20)	u ₉	u ₁₁ (15,24) u ₁₅ (25,25)
		u ₃ (5,21) u ₁₂ (16,22) u ₆ (12,23) u ₁₁ (15,24)
		u ₁₅ (25,25)



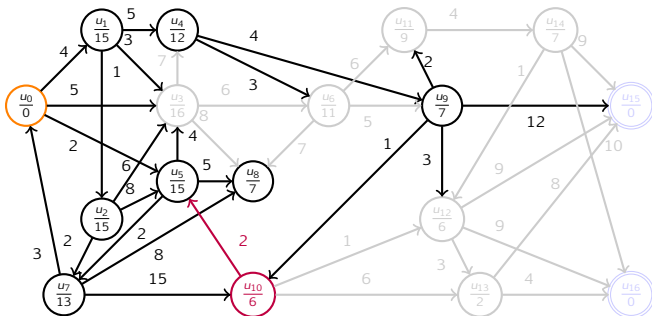
Algorithm A*(u_0, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅ et (u ∉ T)) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

$v = u_5$ $D = \{u_0, u_1, u_2, u_4, u_5, u_7, u_8, u_9, u_{10}\}$

$u(g(u), f(u))$	père(u)	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17) u_1(4,19) u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14) u_7(4,17) u_1(4,19) u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17) u_1(4,19) u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19) u_3(5,21) u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20) u_4(9,21) u_3(5,21) u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21) u_3(5,21) u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20) u_3(5,21) u_6(12,23) u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20) u_3(5,21) u_{12}(16,22) u_6(12,23)$
$u_{10}(14,20)$	u_9	$u_{11}(15,24) u_{15}(25,25)$
		$u_3(5,21) u_{12}(16,22) u_6(12,23) u_{11}(15,24)$
		$u_{15}(25,25)$



Algorithm A*(u_0, Sol)

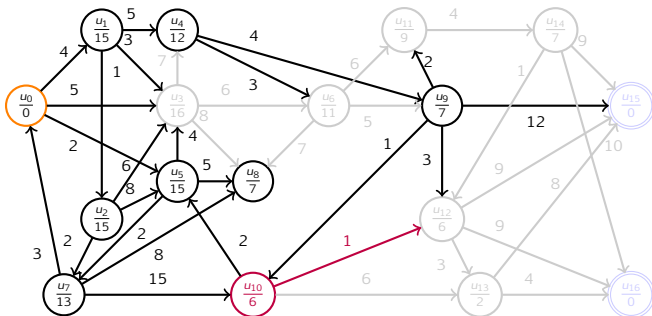
```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅ et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
            v ← suivant(u);
    u ← premier(G);

```

$v = u_{12}$ D = { $u_0, u_1, u_2, u_4, u_5, u_7, u_8, u_9, u_{10}$ }

$u(g(u), f(u))$	père(u)	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17) u_1(4,19) u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14) u_7(4,17) u_1(4,19) u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17) u_1(4,19) u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19) u_3(5,21) u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20) u_4(9,21) u_3(5,21) u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21) u_3(5,21) u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20) u_3(5,21) u_6(12,23) u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20) u_3(5,21) u_{12}(16,22) u_6(12,23)$
$u_{10}(14,20)$	u_9	$u_{11}(15,24) u_{15}(25,25)$
		$u_3(5,21) u_{12}(16,22) u_6(12,23) u_{11}(15,24)$
		$u_{15}(25,25)$



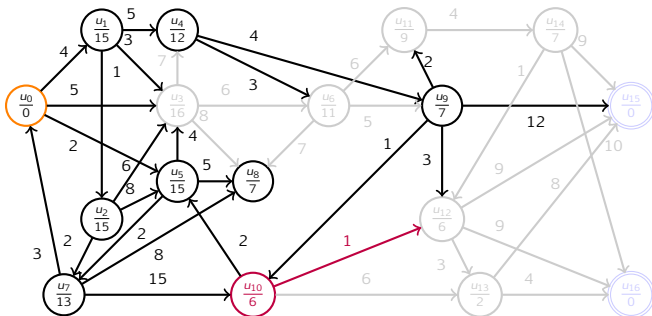
Algorithm A*(u_0, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅ et (u ∉ T)) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

$v = u_{12}$ $D = \{u_0, u_1, u_2, u_4, u_5, u_7, u_8, u_9, u_{10}\}$

$u(g(u), f(u))$	père(u)	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20)u_3(5,21)u_6(12,23)u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20)u_3(5,21)u_{12}(16,22)u_6(12,23)$
$u_{10}(14,20)$	u_9	$u_{11}(15,24)u_{15}(25,25)$
		$u_3(5,21)u_{12}(16,22)u_6(12,23)u_{11}(15,24)$
		$u_{15}(25,25)$



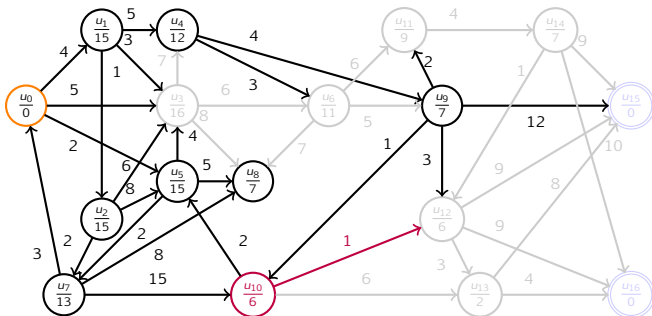
Algorithm A*(u_0, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅ et (u ∉ T)) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

$v = u_{12}$ D={ $u_0, u_1, u_2, u_4, u_5, u_7, u_8, u_9, u_{10}$ }

$u(g(u), f(u))$	père(u)	$G = \{ v \mid v(g(v), f(v)) \}$
$u_0(0,0)$	$\perp$	$u_5(2,17) u_1(4,19) u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14) u_7(4,17) u_1(4,19) u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17) u_1(4,19) u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19) u_3(5,21) u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20) u_4(9,21) u_3(5,21) u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21) u_3(5,21) u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20) u_3(5,21) u_6(12,23) u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20) u_3(5,21) u_{12}(16,22) u_6(12,23)$
$u_{10}(14,20)$	u_9	$u_{11}(15,24) u_{15}(25,25)$
		$u_3(5,21) u_{12}(16,22) u_6(12,23) u_{11}(15,24)$
		$u_{15}(25,25)$



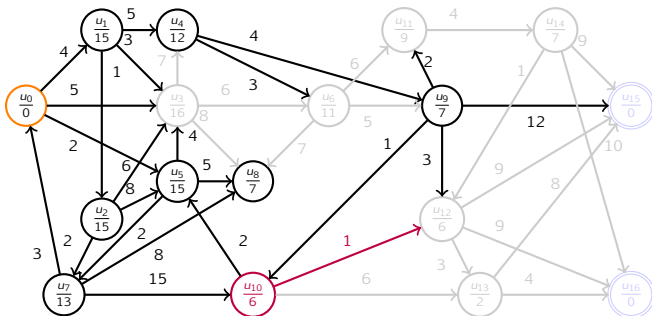
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅ et (u ∉ T)) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

v = u₁₂ D = {u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀}

u(g(u), f(u))	père(u)	G = { v v(g(v), f(v)) }
u ₀ (0,0)	⊥	u ₅ (2,17) u ₁ (4,19) u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14) u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21) u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20) u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20) u ₃ (5,21) u ₆ (12,23) u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20) u ₃ (5,21) u ₁₂ (16,22) u ₆ (12,23)
u ₁₀ (14,20)	u ₉	u ₁₁ (15,24) u ₁₅ (25,25)
		u ₁₂ (15,21) u ₃ (5,21) u ₆ (12,23) u ₁₁ (15,24)
		u ₁₅ (25,25)



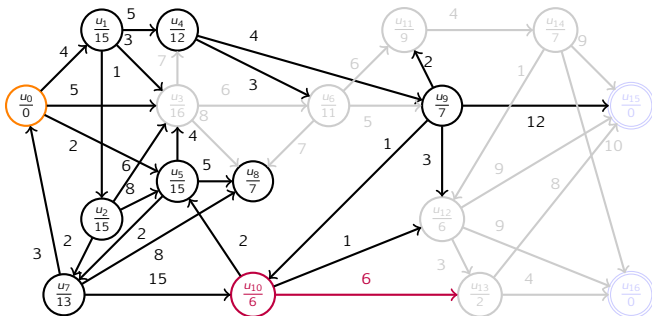
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅ et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
            v ← suivant(u);
    u ← premier(G);
    
```

v = **u₁₃** D = {u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀}

$u(g(u), f(u))$	père(u)	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17) u_1(4,19) u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14) u_7(4,17) u_1(4,19) u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17) u_1(4,19) u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19) u_3(5,21) u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20) u_4(9,21) u_3(5,21) u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21) u_3(5,21) u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20) u_3(5,21) u_6(12,23) u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20) u_3(5,21) u_{12}(16,22) u_6(12,23)$
$u_{10}(14,20)$	u_9	$u_{11}(15,24) u_{15}(25,25)$
		$u_{12}(15,21) u_3(5,21) u_6(12,23) u_{11}(15,24)$
		$u_{15}(25,25)$



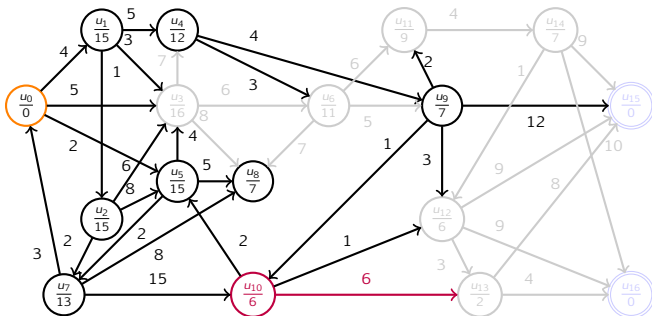
Algorithm A*(u_0, Sol)

```

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    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
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            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

$v = u_{13}$ $D = \{u_0, u_1, u_2, u_4, u_5, u_7, u_8, u_9, u_{10}\}$

$u(g(u), f(u))$	père(u)	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17) u_1(4,19) u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14) u_7(4,17) u_1(4,19) u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17) u_1(4,19) u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19) u_3(5,21) u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20) u_4(9,21) u_3(5,21) u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21) u_3(5,21) u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20) u_3(5,21) u_6(12,23) u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20) u_3(5,21) u_{12}(16,22) u_6(12,23)$
$u_{10}(14,20)$	u_9	$u_{11}(15,24) u_{15}(25,25)$
		$u_{12}(15,21) u_3(5,21) u_6(12,23) u_{11}(15,24)$
		$u_{15}(25,25)$



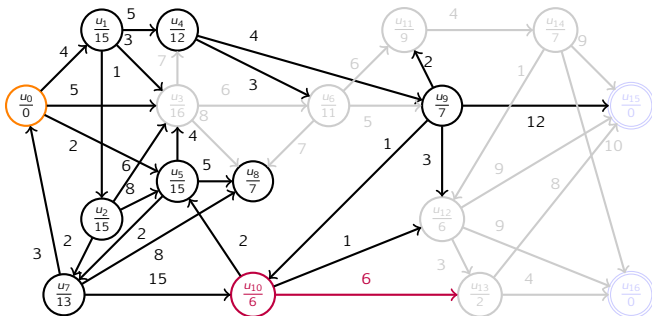
Algorithm A*(u_0, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
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    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
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            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

$v = u_{13}$ $D = \{u_0, u_1, u_2, u_4, u_5, u_7, u_8, u_9, u_{10}\}$

$u(g(u), f(u))$	$\text{père}(u)$	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17) u_1(4,19) u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14) u_7(4,17) u_1(4,19) u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17) u_1(4,19) u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19) u_3(5,21) u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20) u_4(9,21) u_3(5,21) u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21) u_3(5,21) u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20) u_3(5,21) u_6(12,23) u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20) u_3(5,21) u_{12}(16,22) u_6(12,23)$
$u_{10}(14,20)$	u_9	$u_{11}(15,24) u_{15}(25,25)$
		$u_{12}(15,21) u_3(5,21) u_6(12,23) u_{11}(15,24)$
		$u_{15}(25,25)$



Algorithm A*(u_0, Sol)

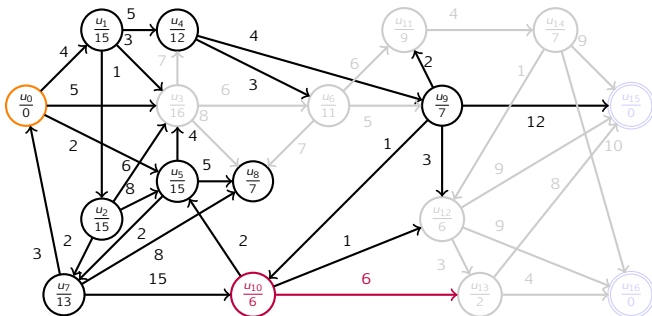
```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅ et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);

```

$v = u_{13}$ $D = \{u_0, u_1, u_2, u_4, u_5, u_7, u_8, u_9, u_{10}\}$

$u(g(u), f(u))$	père(u)	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17) u_1(4,19) u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14) u_7(4,17) u_1(4,19) u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17) u_1(4,19) u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19) u_3(5,21) u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20) u_4(9,21) u_3(5,21) u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21) u_3(5,21) u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20) u_3(5,21) u_6(12,23) u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20) u_3(5,21) u_{12}(16,22) u_6(12,23)$
$u_{10}(14,20)$	u_9	$u_{11}(15,24) u_{15}(25,25)$
		$u_{12}(15,21) u_3(5,21) u_{13}(20,22) u_6(12,23)$
		$u_{11}(15,24) u_{15}(25,25)$



Algorithm A*(u₀, Sol)

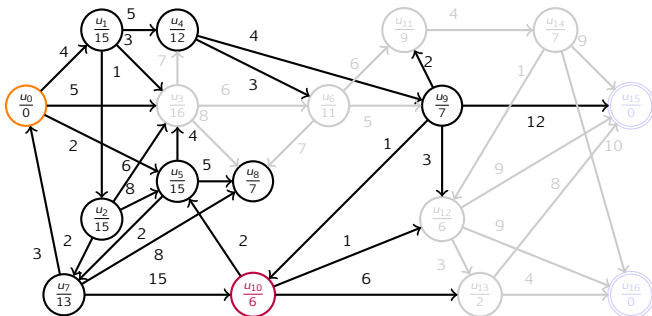
```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅ et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
            v ← suivant(u);
    u ← premier(G);

```

v = $\perp$ D = {u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀}

u(g(u), f(u))	père(u)	G = { v v(g(v), f(v)) }
u ₀ (0,0)	$\perp$	u ₅ (2,17) u ₁ (4,19) u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14) u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21) u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20) u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20) u ₃ (5,21) u ₆ (12,23) u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20) u ₃ (5,21) u ₁₂ (16,22) u ₆ (12,23)
u ₁₀ (14,20)	u ₉	u ₁₁ (15,24) u ₁₅ (25,25)
		u ₁₂ (15,21) u ₃ (5,21) u ₁₃ (20,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)



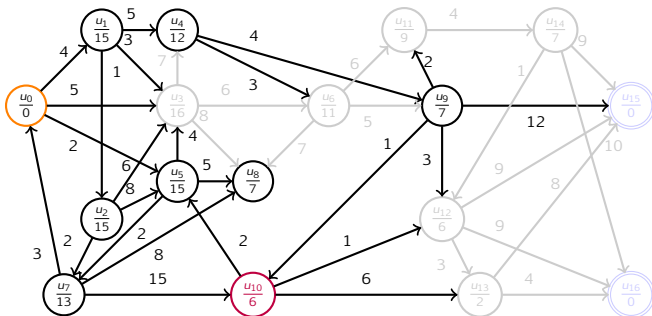
Algorithm A*(u_0, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅ et (u ∉ T)) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

$v \in D = \{u_0, u_1, u_2, u_4, u_5, u_7, u_8, u_9, u_{10}\}$

$u(g(u), f(u))$	père(u)	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20)u_3(5,21)u_6(12,23)u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20)u_3(5,21)u_{12}(16,22)u_6(12,23)$
$u_{10}(14,20)$	u_9	$u_{11}(15,24)u_{15}(25,25)$
		$u_{12}(15,21)u_3(5,21)u_{13}(20,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$



Algorithm A*(u₀, Sol)

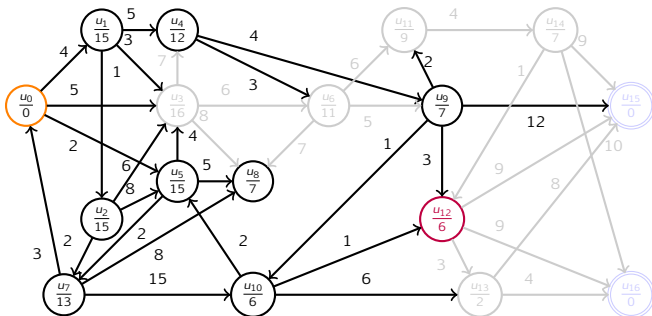
```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);

```

v = 1 D = {u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀}

u(g(u), f(u))	père(u)	G = { v v(g(v), f(v)) }
u ₀ (0,0)	⊥	u ₅ (2,17) u ₁ (4,19) u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14) u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21) u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20) u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20) u ₃ (5,21) u ₆ (12,23) u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20) u ₃ (5,21) u ₁₂ (16,22) u ₆ (12,23)
u ₁₀ (14,20)	u ₉	u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₂ (15,21)	u ₁₀	u ₁₂ (15,21) u ₃ (5,21) u ₁₃ (20,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)



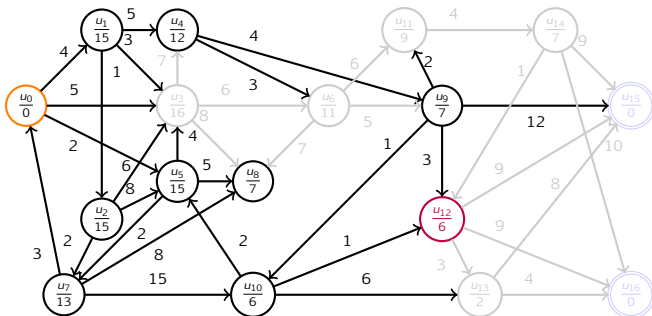
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅ et (u ∉ T)) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

v = 1 D = {u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀}

u(g(u),f(u))	père(u)	G = { v v(g(v),f(v)) }
u ₀ (0,0)	⊥	u ₅ (2,17) u ₁ (4,19) u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14) u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21) u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20) u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20) u ₃ (5,21) u ₆ (12,23) u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20) u ₃ (5,21) u ₁₂ (16,22) u ₆ (12,23)
u ₁₀ (14,20)	u ₉	u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₂ (15,21)	u ₁₀	u ₁₂ (15,21) u ₃ (5,21) u ₁₃ (20,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)



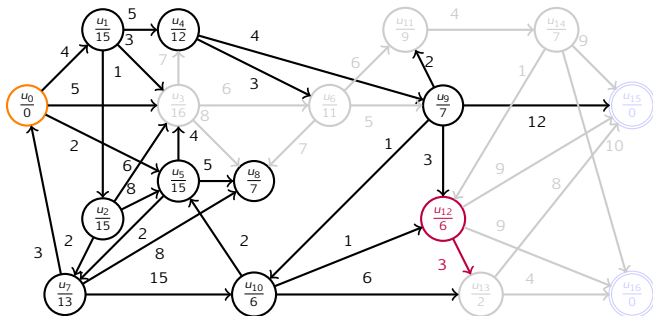
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

v = **u₁₂** D = {u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀, **u₁₂**}

u(g(u), f(u))	père(u)	G = { v v(g(v), f(v)) }
u ₀ (0,0)	⊥	u ₅ (2,17) u ₁ (4,19) u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14) u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21) u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20) u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20) u ₃ (5,21) u ₆ (12,23) u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20) u ₃ (5,21) u ₁₂ (16,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₀ (14,20)	u ₉	u ₁₂ (15,21) u ₃ (5,21) u ₁₃ (20,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₂ (15,21)	u ₁₀	u ₃ (5,21) u ₆ (12,23) u ₁₃ (20,22)
		u ₁₁ (15,24) u ₁₅ (25,25)



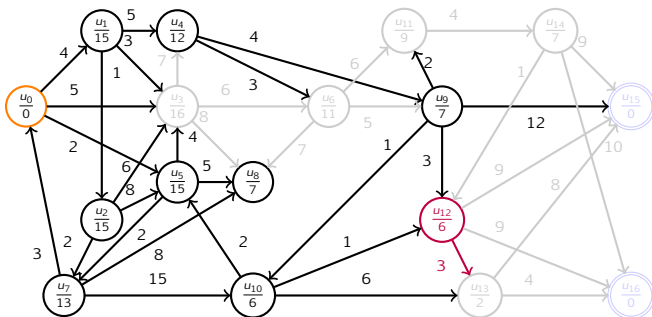
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

v = u₁₃ D = {u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀, u₁₂}

$u(g(u), f(u))$	$père(u)$	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20)u_3(5,21)u_6(12,23)u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20)u_3(5,21)u_{12}(16,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{10}(14,20)$	u_9	$u_{12}(15,21)u_3(5,21)u_{13}(20,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{12}(15,21)$	u_{10}	$u_3(5,21)u_6(12,23)u_{13}(20,22)$
		$u_{11}(15,24)u_{15}(25,25)$



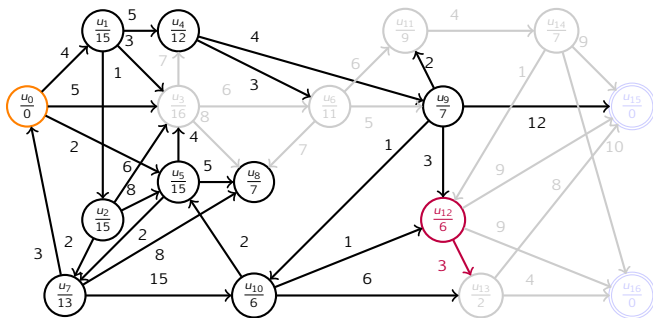
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

v=u₁₃ D={u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀, u₁₂}

u(g(u),f(u))	père(u)	G = { v v(g(v),f(v)) }
u ₀ (0,0)	⊥	u ₅ (2,17)u ₁ (4,19)u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14)u ₇ (4,17)u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21)u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20)u ₄ (9,21)u ₃ (5,21)u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21)u ₃ (5,21)u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20)u ₃ (5,21)u ₆ (12,23)u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20)u ₃ (5,21)u ₁₂ (16,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₀ (14,20)	u ₉	u ₁₂ (15,21)u ₃ (5,21)u ₁₃ (20,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₂ (15,21)	u ₁₀	u ₃ (5,21)u ₆ (12,23)u ₁₃ (20,22)
		u ₁₁ (15,24)u ₁₅ (25,25)



Algorithm A*(u₀, Sol)

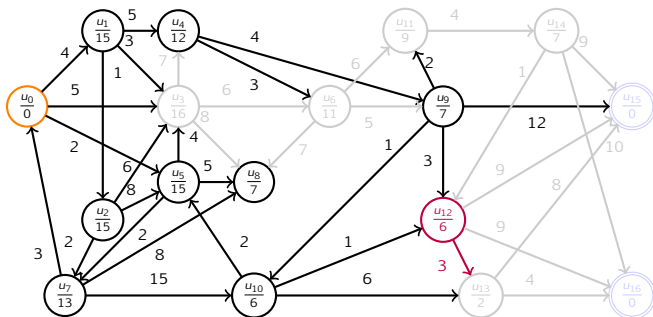
```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);

```

v = u₁₃ D = {u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀, u₁₂}

u(g(u), f(u))	père(u)	G = { v v(g(v), f(v)) }
u ₀ (0,0)	⊥	u ₅ (2,17) u ₁ (4,19) u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14) u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21) u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20) u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20) u ₃ (5,21) u ₆ (12,23) u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20) u ₃ (5,21) u ₁₂ (16,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₀ (14,20)	u ₉	u ₁₂ (15,21) u ₃ (5,21) u ₁₃ (20,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₂ (15,21)	u ₁₀	u₁₃(18,20) u ₃ (5,21) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)



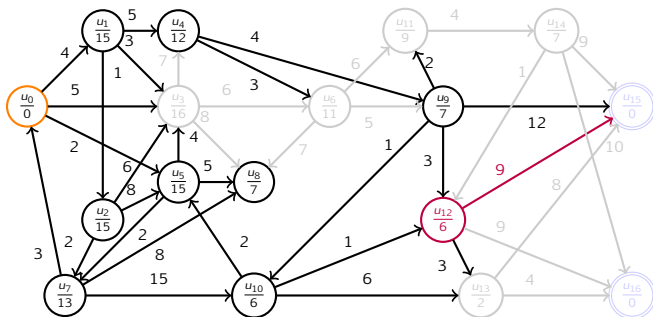
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
            v ← suivant(u);
    u ← premier(G);
    
```

v = **u₁₅** D = {u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀, u₁₂}

u(g(u), f(u))	père(u)	G = { v v(g(v), f(v)) }
u ₀ (0,0)	⊥	u ₅ (2,17) u ₁ (4,19) u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14) u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21) u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20) u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20) u ₃ (5,21) u ₆ (12,23) u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20) u ₃ (5,21) u ₁₂ (16,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₀ (14,20)	u ₉	u ₁₂ (15,21) u ₃ (5,21) u ₁₃ (20,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₂ (15,21)	u ₁₀	u ₁₃ (18,20) u ₃ (5,21) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)



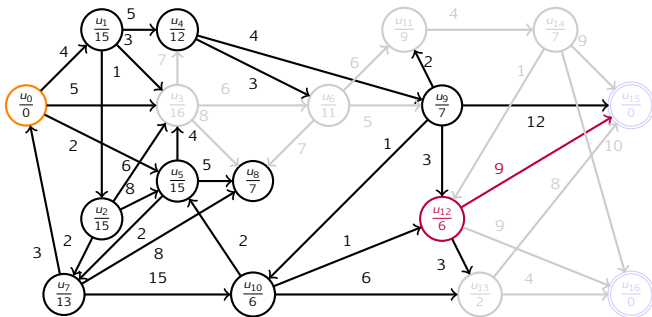
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

v=u₁₅ D={u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀, u₁₂}

u(g(u),f(u))	père(u)	G = { v v(g(v),f(v)) }
u ₀ (0,0)	⊥	u ₅ (2,17)u ₁ (4,19)u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14)u ₇ (4,17)u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21)u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20)u ₄ (9,21)u ₃ (5,21)u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21)u ₃ (5,21)u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20)u ₃ (5,21)u ₆ (12,23)u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20)u ₃ (5,21)u ₁₂ (16,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₀ (14,20)	u ₉	u ₁₂ (15,21)u ₃ (5,21)u ₁₃ (20,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₂ (15,21)	u ₁₀	u ₁₃ (18,20)u ₃ (5,21)u ₆ (12,23)
		u ₁₁ (15,24)u ₁₅ (25,25)



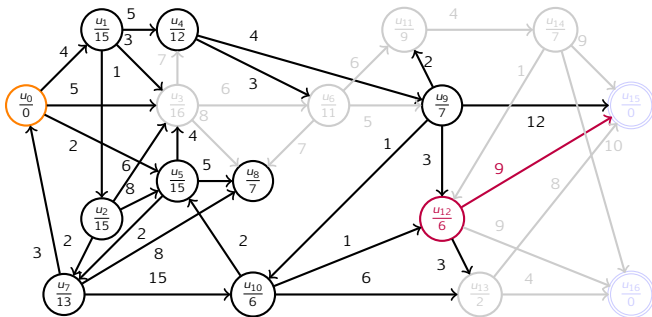
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

v=u₁₅ D={u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀, u₁₂}

u(g(u),f(u))	père(u)	G = { v v(g(v),f(v)) }
u ₀ (0,0)	⊥	u ₅ (2,17)u ₁ (4,19)u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14)u ₇ (4,17)u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21)u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20)u ₄ (9,21)u ₃ (5,21)u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21)u ₃ (5,21)u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20)u ₃ (5,21)u ₆ (12,23)u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20)u ₃ (5,21)u ₁₂ (16,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₀ (14,20)	u ₉	u ₁₂ (15,21)u ₃ (5,21)u ₁₃ (20,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₂ (15,21)	u ₁₀	u ₁₃ (18,20)u ₃ (5,21)u ₆ (12,23)
		u ₁₁ (15,24)u ₁₅ (25,25)



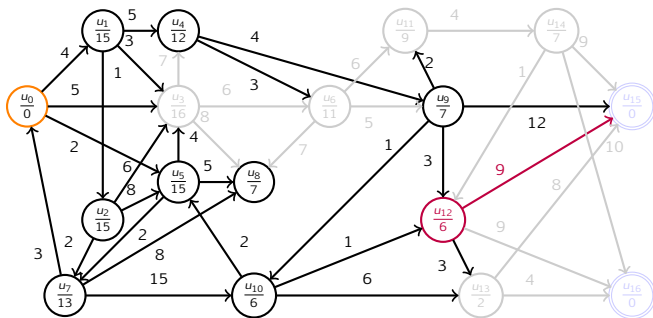
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

v=u₁₅ D={u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀, u₁₂}

u(g(u),f(u))	père(u)	G = { v v(g(v),f(v)) }
u ₀ (0,0)	⊥	u ₅ (2,17)u ₁ (4,19)u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14)u ₇ (4,17)u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21)u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20)u ₄ (9,21)u ₃ (5,21)u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21)u ₃ (5,21)u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20)u ₃ (5,21)u ₆ (12,23)u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20)u ₃ (5,21)u ₁₂ (16,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₀ (14,20)	u ₉	u ₁₂ (15,21)u ₃ (5,21)u ₁₃ (20,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₂ (15,21)	u ₁₀	u ₁₃ (18,20)u ₃ (5,21)u ₆ (12,23) u₁₅(24,24)
		u ₁₁ (15,24)



Algorithme $A^*(u_0, \text{Sol})$

$$G \leftarrow \{u_0\}; D \leftarrow \emptyset; g(u_0) \leftarrow 0; f(u_0) \leftarrow 0; u \leftarrow \text{premier}(G);$$

```
while ( $G \neq \emptyset$ ) et ( $u \notin T$ ) do
```

$$G \leftarrow G \setminus \{u\}; D \leftarrow D \cup \{u\}; v \leftarrow \text{suivant}(u);$$

```
while (v est valide) do
```

if $((v \notin (D \cup G))$ ou $(g(v) > g(u) + k(u,v))$ then

$$g(v) \leftarrow g(u) + k(u,v); f(v) \leftarrow g(v) + h(v);$$

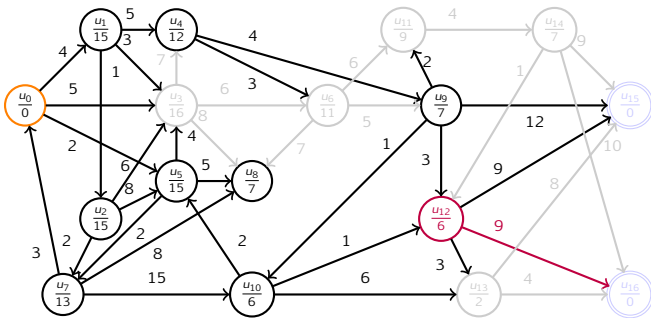
père(v) ← u ; insérer-selon-f(G,v);

vivant(u).

```
v ← suivant(u);
```

```
u ← premier(G);
```

$u(g(u), f(u))$	$p\grave{e}re(u)$	$G = \{ v \mid v(g(v), f(v)) \}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20)u_3(5,21)u_6(12,23)u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20)u_3(5,21)u_{12}(16,22)u_6(12,23)$ $u_{11}(15,24)u_{15}(25,25)$
$u_{10}(14,20)$	u_9	$u_{12}(15,21)u_3(5,21)u_{13}(20,22)u_6(12,23)$ $u_{11}(15,24)u_{15}(25,25)$
$u_{12}(15,21)$	u_{10}	$u_{13}(18,20)u_3(5,21)u_6(12,23)u_{15}(24,24)$ $u_{11}(15,24)$

$$v = u_{16} \quad D = \{u_0, u_1, u_2, u_4, u_5, u_7, u_8, u_9, u_{10}, u_{12}\}$$


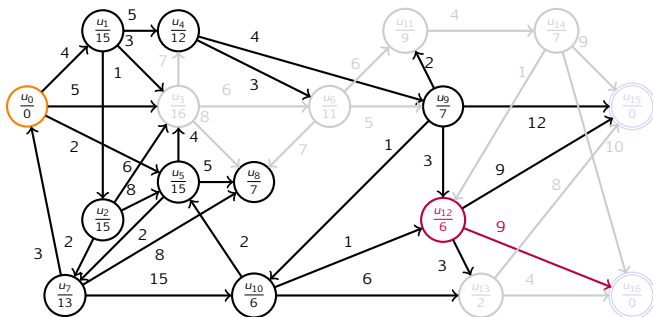
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

v=u₁₆ D={u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀, u₁₂}

u(g(u),f(u))	père(u)	G = { v v(g(v),f(v)) }
u ₀ (0,0)	⊥	u ₅ (2,17)u ₁ (4,19)u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14)u ₇ (4,17)u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21)u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20)u ₄ (9,21)u ₃ (5,21)u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21)u ₃ (5,21)u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20)u ₃ (5,21)u ₆ (12,23)u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20)u ₃ (5,21)u ₁₂ (16,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₀ (14,20)	u ₉	u ₁₂ (15,21)u ₃ (5,21)u ₁₃ (20,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₂ (15,21)	u ₁₀	u ₁₃ (18,20)u ₃ (5,21)u ₆ (12,23) u ₁₅ (24,24)
		u ₁₁ (15,24)



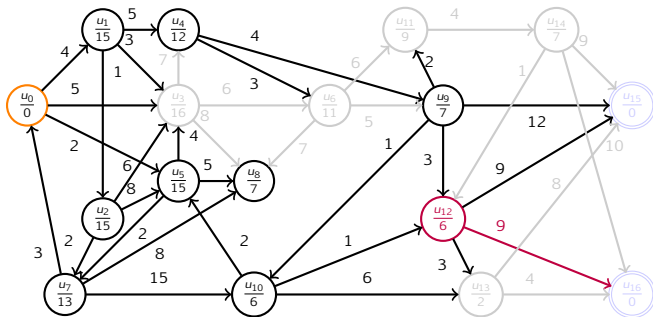
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
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        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
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            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

v=u₁₆ D={u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀, u₁₂}

u(g(u),f(u))	père(u)	G = { v v(g(v),f(v)) }
u ₀ (0,0)	⊥	u ₅ (2,17)u ₁ (4,19)u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14)u ₇ (4,17)u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21)u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20)u ₄ (9,21)u ₃ (5,21)u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21)u ₃ (5,21)u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20)u ₃ (5,21)u ₆ (12,23)u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20)u ₃ (5,21)u ₁₂ (16,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₀ (14,20)	u ₉	u ₁₂ (15,21)u ₃ (5,21)u ₁₃ (20,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₂ (15,21)	u ₁₀	u ₁₃ (18,20)u ₃ (5,21)u ₆ (12,23) u ₁₅ (24,24)
		u ₁₁ (15,24)



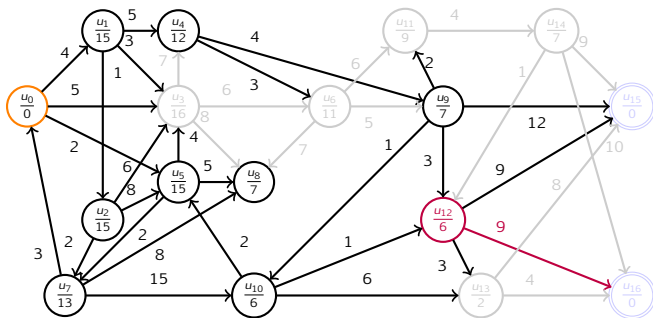
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
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        v ← suivant(u);
    u ← premier(G);
    
```

v=u₁₆ D={u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀, u₁₂}

u(g(u),f(u))	père(u)	G = { v v(g(v),f(v)) }
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u ₅ (2,17)	u ₀	u ₈ (7,14)u ₇ (4,17)u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21)u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20)u ₄ (9,21)u ₃ (5,21)u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21)u ₃ (5,21)u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20)u ₃ (5,21)u ₆ (12,23)u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20)u ₃ (5,21)u ₁₂ (16,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₀ (14,20)	u ₉	u ₁₂ (15,21)u ₃ (5,21)u ₁₃ (20,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₂ (15,21)	u ₁₀	u ₁₃ (18,20)u ₃ (5,21)u ₆ (12,23) u ₁₅ (24,24)
		u₁₆(24,24) u ₁₁ (15,24)



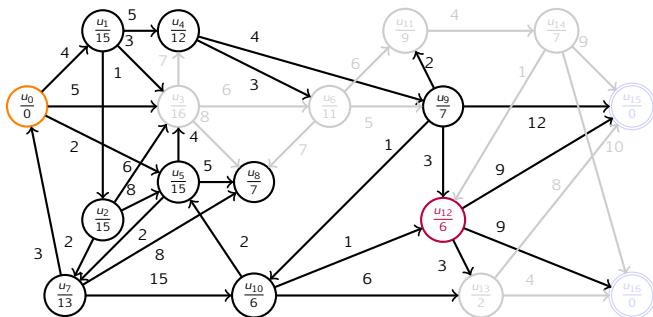
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

v = ⊥ D = {u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀, u₁₂}

u(g(u), f(u))	père(u)	G = { v v(g(v), f(v)) }
u ₀ (0,0)	⊥	u ₅ (2,17) u ₁ (4,19) u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14) u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21) u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20) u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20) u ₃ (5,21) u ₆ (12,23) u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20) u ₃ (5,21) u ₁₂ (16,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₀ (14,20)	u ₉	u ₁₂ (15,21) u ₃ (5,21) u ₁₃ (20,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₂ (15,21)	u ₁₀	u ₁₃ (18,20) u ₃ (5,21) u ₆ (12,23) u ₁₅ (24,24)
		u ₁₆ (24,24) u ₁₁ (15,24)



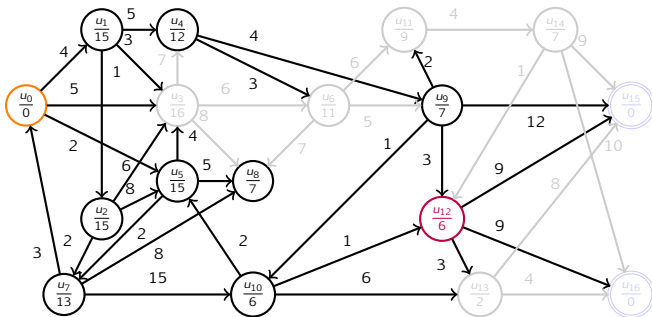
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

v = ⊥ D = {u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀, u₁₂}

u(g(u), f(u))	père(u)	G = { v v(g(v), f(v)) }
u ₀ (0,0)	⊥	u ₅ (2,17) u ₁ (4,19) u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14) u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21) u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20) u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20) u ₃ (5,21) u ₆ (12,23) u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20) u ₃ (5,21) u ₁₂ (16,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₀ (14,20)	u ₉	u ₁₂ (15,21) u ₃ (5,21) u ₁₃ (20,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₂ (15,21)	u ₁₀	u ₁₃ (18,20) u ₃ (5,21) u ₆ (12,23) u ₁₅ (24,24)
		u ₁₆ (24,24) u ₁₁ (15,24)



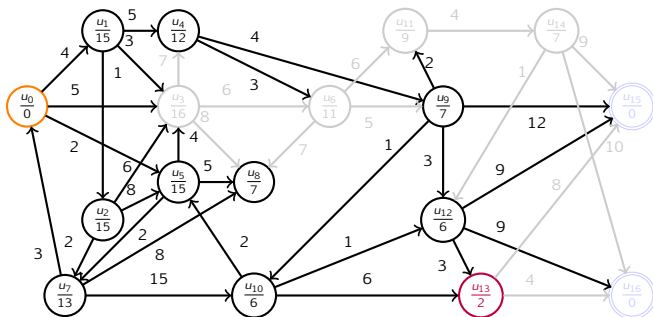
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

v ← ∅ D ← {u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀, u₁₂}

u(g(u), f(u))	père(u)	G = { v v(g(v), f(v)) }
u ₀ (0,0)	⊥	u ₅ (2,17) u ₁ (4,19) u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14) u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21) u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20) u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20) u ₃ (5,21) u ₆ (12,23) u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20) u ₃ (5,21) u ₁₂ (16,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₀ (14,20)	u ₉	u ₁₂ (15,21) u ₃ (5,21) u ₁₃ (20,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₂ (15,21)	u ₁₀	u ₁₃ (18,20) u ₃ (5,21) u ₆ (12,23) u ₁₅ (24,24)
u ₁₃ (18,20)	u ₁₂	u ₁₆ (24,24) u ₁₁ (15,24)



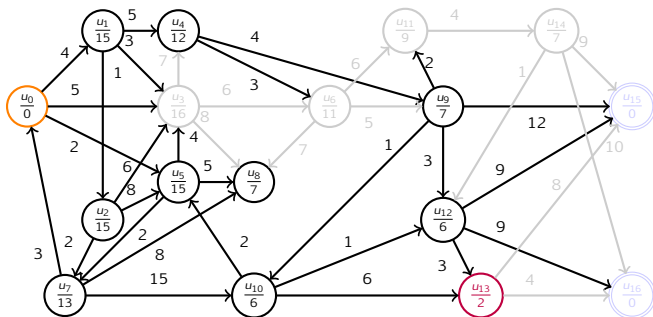
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

v ← ∅ D ← {u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀, u₁₂}

u(g(u), f(u))	père(u)	G = { v v(g(v), f(v)) }
u ₀ (0,0)	⊥	u ₅ (2,17) u ₁ (4,19) u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14) u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21) u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20) u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20) u ₃ (5,21) u ₆ (12,23) u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20) u ₃ (5,21) u ₁₂ (16,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₀ (14,20)	u ₉	u ₁₂ (15,21) u ₃ (5,21) u ₁₃ (20,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₂ (15,21)	u ₁₀	u ₁₃ (18,20) u ₃ (5,21) u ₆ (12,23) u ₁₅ (24,24)
u ₁₃ (18,20)	u ₁₂	u ₁₆ (24,24) u ₁₁ (15,24)



Algorithm A*(u₀, Sol)

$G \leftarrow \{u_0\}$; $D \leftarrow \emptyset$; $g(u_0) \leftarrow 0$; $f(u_0) \leftarrow 0$; $u \leftarrow \text{premier}(G)$;

while ($G \neq \emptyset$) **et** ($u \notin T$) **do**

$G \leftarrow G \setminus \{u\}$; $D \leftarrow D \cup \{u\}$; $v \leftarrow \text{suivant}(u)$;

while (v est valide) **do**

if ($(v \notin (D \cup G))$ **ou** ($g(v) > g(u) + k(u, v)$)) **then**

$g(v) \leftarrow g(u) + k(u, v)$; $f(v) \leftarrow g(v) + h(v)$;

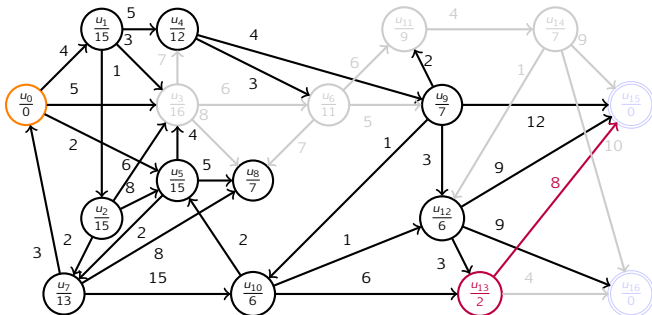
$\text{père}(v) \leftarrow u$; insérer-selon- $f(G, v)$;

$v \leftarrow \text{suivant}(u)$;

$u \leftarrow \text{premier}(G)$;

$v = u_{15}$ $D = \{u_0, u_1, u_2, u_4, u_5, u_7, u_8, u_9, u_{10}, u_{12}, u_{13}\}$

$u(g(u), f(u))$	$\text{père}(u)$	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20)u_3(5,21)u_6(12,23)u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20)u_3(5,21)u_{12}(16,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{10}(14,20)$	u_9	$u_{12}(15,21)u_3(5,21)u_{13}(20,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{12}(15,21)$	u_{10}	$u_{13}(18,20)u_3(5,21)u_6(12,23)u_{15}(24,24)$
		$u_{16}(24,24)u_{11}(15,24)$
$u_{13}(18,20)$	u_{12}	$u_3(5,21)u_6(12,23)u_{15}(24,24)u_{16}(24,24)$
		$u_{11}(15,24)$



Algorithm A*(u₀, Sol)

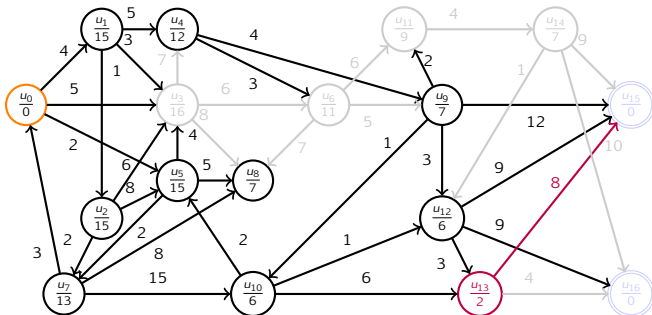
```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);

```

v = u₁₅ D = {u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀, u₁₂, u₁₃}

u(g(u), f(u))	père(u)	G = { v v(g(v), f(v)) }
u ₀ (0,0)	⊥	u ₅ (2,17) u ₁ (4,19) u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14) u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21) u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20) u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20) u ₃ (5,21) u ₆ (12,23) u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20) u ₃ (5,21) u ₁₂ (16,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₀ (14,20)	u ₉	u ₁₂ (15,21) u ₃ (5,21) u ₁₃ (20,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₂ (15,21)	u ₁₀	u ₁₃ (18,20) u ₃ (5,21) u ₆ (12,23) u ₁₅ (24,24)
		u ₁₆ (24,24) u ₁₁ (15,24)
u ₁₃ (18,20)	u ₁₂	u ₃ (5,21) u ₆ (12,23) u ₁₅ (24,24) u ₁₆ (24,24)
		u ₁₁ (15,24)

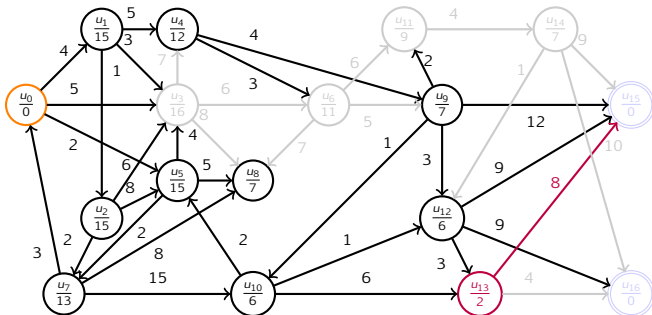


Algorithm A*(u₀, Sol)

$G \leftarrow \{u_0\}$; $D \leftarrow \emptyset$; $g(u_0) \leftarrow 0$; $f(u_0) \leftarrow 0$; $u \leftarrow \text{premier}(G)$;
while ($G \neq \emptyset$) **et** ($u \notin T$) **do**
 $G \leftarrow G \setminus \{u\}$; $D \leftarrow D \cup \{u\}$; $v \leftarrow \text{suivant}(u)$;
 while (v est valide) **do**
 if ($(v \notin (D \cup G))$ **ou** ($g(v) > g(u) + k(u,v)$)) **then**
 $g(v) \leftarrow g(u) + k(u,v)$; $f(v) \leftarrow g(v) + h(v)$;
 $\text{père}(v) \leftarrow u$; insérer-selon- $f(G,v)$;
 $v \leftarrow \text{suivant}(u)$;
 $u \leftarrow \text{premier}(G)$;
until ($u \in T$);

$v = u_{15}$ $D = \{u_0, u_1, u_2, u_4, u_5, u_7, u_8, u_9, u_{10}, u_{12}, u_{13}\}$

$u(g(u), f(u))$	$\text{père}(u)$	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20)u_3(5,21)u_6(12,23)u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20)u_3(5,21)u_{12}(16,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{10}(14,20)$	u_9	$u_{12}(15,21)u_3(5,21)u_{13}(20,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{12}(15,21)$	u_{10}	$u_{13}(18,20)u_3(5,21)u_6(12,23)u_{15}(24,24)$
		$u_{16}(24,24)u_{11}(15,24)$
$u_{13}(18,20)$	u_{12}	$u_3(5,21)u_6(12,23)u_{15}(24,24)u_{16}(24,24)$
		$u_{11}(15,24)$



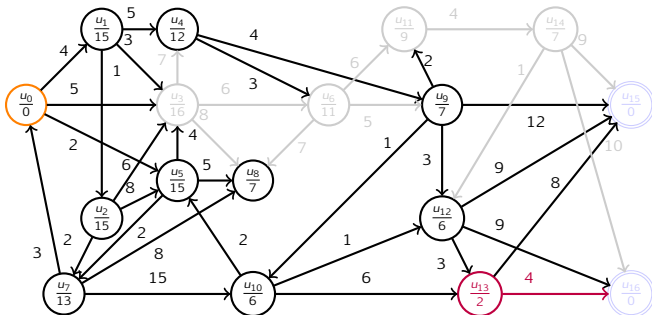
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∉ T) do
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        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
            v ← suivant(u);
    u ← premier(G);
    
```

v = **u₁₆** D = {u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀, u₁₂, u₁₃}

u(g(u), f(u))	père(u)	G = { v v(g(v), f(v)) }
u ₀ (0,0)	⊥	u ₅ (2,17) u ₁ (4,19) u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14) u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21) u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20) u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20) u ₃ (5,21) u ₆ (12,23) u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20) u ₃ (5,21) u ₁₂ (16,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₀ (14,20)	u ₉	u ₁₂ (15,21) u ₃ (5,21) u ₁₃ (20,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₂ (15,21)	u ₁₀	u ₁₃ (18,20) u ₃ (5,21) u ₆ (12,23) u ₁₅ (24,24)
		u ₁₆ (24,24) u ₁₁ (15,24)
u ₁₃ (18,20)	u ₁₂	u ₃ (5,21) u ₆ (12,23) u ₁₅ (24,24) u ₁₆ (24,24)
		u ₁₁ (15,24)



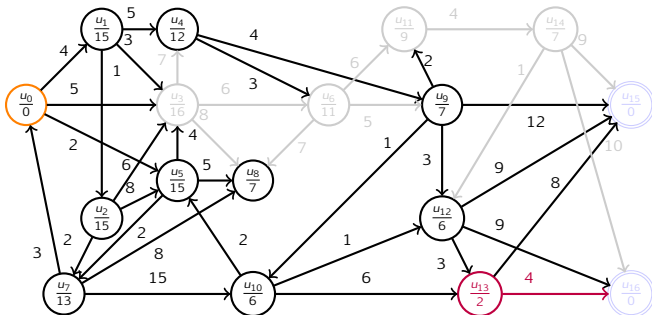
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∉ T) do
  G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
  while (v est valide) do
    if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
      g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
      père(v) ← u; insérer-selon-f(G,v);
    v ← suivant(u);
  u ← premier(G);
  
```

v=u₁₆ D={u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀, u₁₂, u₁₃}

u(g(u),f(u))	père(u)	G = { v v(g(v),f(v)) }
u ₀ (0,0)	⊥	u ₅ (2,17)u ₁ (4,19)u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14)u ₇ (4,17)u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21)u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20)u ₄ (9,21)u ₃ (5,21)u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21)u ₃ (5,21)u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20)u ₃ (5,21)u ₆ (12,23)u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20)u ₃ (5,21)u ₁₂ (16,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₀ (14,20)	u ₉	u ₁₂ (15,21)u ₃ (5,21)u ₁₃ (20,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₂ (15,21)	u ₁₀	u ₁₃ (18,20)u ₃ (5,21)u ₆ (12,23) u ₁₅ (24,24)
		u ₁₆ (24,24)u ₁₁ (15,24)
u ₁₃ (18,20)	u ₁₂	u ₃ (5,21) u ₆ (12,23)u ₁₅ (24,24) u ₁₆ (24,24)
		u ₁₁ (15,24)



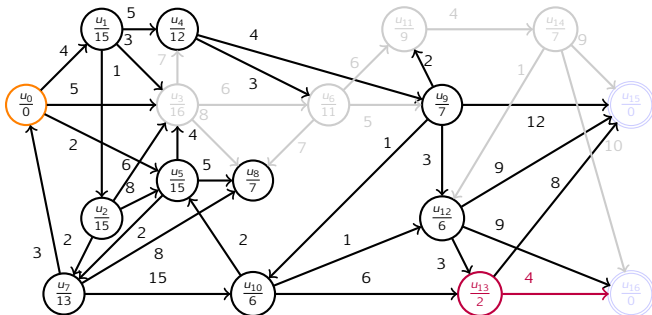
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

v=u₁₆ D={u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀, u₁₂, u₁₃}

u(g(u),f(u))	père(u)	G = { v v(g(v),f(v)) }
u ₀ (0,0)	⊥	u ₅ (2,17)u ₁ (4,19)u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14)u ₇ (4,17)u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21)u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20)u ₄ (9,21)u ₃ (5,21)u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21)u ₃ (5,21)u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20)u ₃ (5,21)u ₆ (12,23)u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20)u ₃ (5,21)u ₁₂ (16,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₀ (14,20)	u ₉	u ₁₂ (15,21)u ₃ (5,21)u ₁₃ (20,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₂ (15,21)	u ₁₀	u ₁₃ (18,20)u ₃ (5,21)u ₆ (12,23) u ₁₅ (24,24)
		u ₁₆ (24,24)u ₁₁ (15,24)
u ₁₃ (18,20)	u ₁₂	u ₃ (5,21) u ₆ (12,23)u ₁₅ (24,24) u ₁₆ (24,24)
		u ₁₁ (15,24)



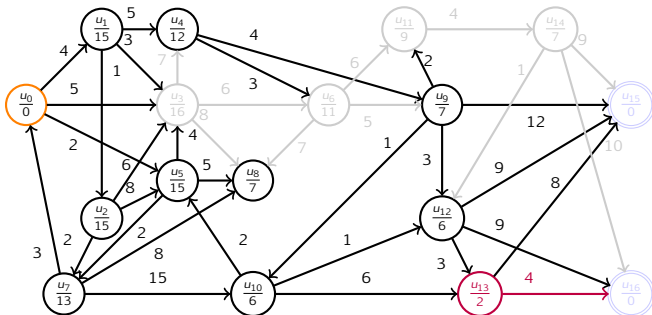
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);
    
```

v=u₁₆ D={u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀, u₁₂, u₁₃}

u(g(u),f(u))	père(u)	G = { v v(g(v),f(v)) }
u ₀ (0,0)	⊥	u ₅ (2,17)u ₁ (4,19)u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14)u ₇ (4,17)u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21)u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20)u ₄ (9,21)u ₃ (5,21)u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21)u ₃ (5,21)u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20)u ₃ (5,21)u ₆ (12,23)u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20)u ₃ (5,21)u ₁₂ (16,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₀ (14,20)	u ₉	u ₁₂ (15,21)u ₃ (5,21)u ₁₃ (20,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₂ (15,21)	u ₁₀	u ₁₃ (18,20)u ₃ (5,21)u ₆ (12,23) u ₁₅ (24,24)
		u ₁₆ (24,24)u ₁₁ (15,24)
u ₁₃ (18,20)	u ₁₂	u ₃ (5,21) u ₁₆ (22,22)u ₆ (12,23)u ₁₅ (24,24)
		u ₁₁ (15,24)



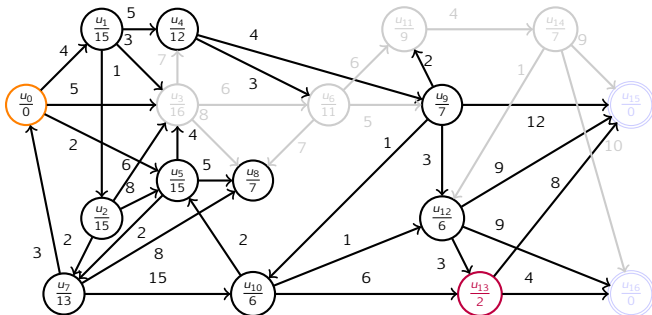
Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
            v ← suivant(u);
    u ← premier(G);
    
```

v = ⊥ D = {u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀, u₁₂, u₁₃}

u(g(u), f(u))	père(u)	G = { v v(g(v), f(v)) }
u ₀ (0,0)	⊥	u ₅ (2,17) u ₁ (4,19) u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14) u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21) u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20) u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20) u ₃ (5,21) u ₆ (12,23) u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20) u ₃ (5,21) u ₁₂ (16,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₀ (14,20)	u ₉	u ₁₂ (15,21) u ₃ (5,21) u ₁₃ (20,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₂ (15,21)	u ₁₀	u ₁₃ (18,20) u ₃ (5,21) u ₆ (12,23) u ₁₅ (24,24)
		u ₁₆ (24,24) u ₁₁ (15,24)
u ₁₃ (18,20)	u ₁₂	u ₃ (5,21) u ₁₆ (22,22) u ₆ (12,23) u ₁₅ (24,24)
		u ₁₁ (15,24)



Algorithm A*(u₀, Sol)

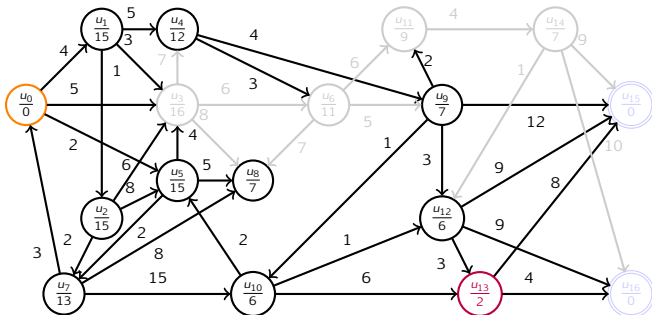
```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∉ T) do
    G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
    while (v est valide) do
        if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
            g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
            père(v) ← u; insérer-selon-f(G,v);
        v ← suivant(u);
    u ← premier(G);

```

v = ⊥ D = {u₀, u₁, u₂, u₄, u₅, u₇, u₈, u₉, u₁₀, u₁₂, u₁₃}

u(g(u), f(u))	père(u)	G = { v v(g(v), f(v)) }
u ₀ (0,0)	⊥	u ₅ (2,17) u ₁ (4,19) u ₃ (5,21)
u ₅ (2,17)	u ₀	u ₈ (7,14) u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₈ (7,14)	u ₅	u ₇ (4,17) u ₁ (4,19) u ₃ (5,21)
u ₇ (4,17)	u ₅	u ₁ (4,19) u ₃ (5,21) u ₁₀ (19,25)
u ₁ (4,19)	u ₀	u ₂ (5,20) u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₂ (5,20)	u ₁	u ₄ (9,21) u ₃ (5,21) u ₁₀ (19,25)
u ₄ (9,21)	u ₁	u ₉ (13,20) u ₃ (5,21) u ₆ (12,23) u ₁₀ (19,25)
u ₉ (13,20)	u ₄	u ₁₀ (14,20) u ₃ (5,21) u ₁₂ (16,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₀ (14,20)	u ₉	u ₁₂ (15,21) u ₃ (5,21) u ₁₃ (20,22) u ₆ (12,23)
		u ₁₁ (15,24) u ₁₅ (25,25)
u ₁₂ (15,21)	u ₁₀	u ₁₃ (18,20) u ₃ (5,21) u ₆ (12,23) u ₁₅ (24,24)
		u ₁₆ (24,24) u ₁₁ (15,24)
u ₁₃ (18,20)	u ₁₂	u ₃ (5,21) u ₁₆ (22,22) u ₆ (12,23) u ₁₅ (24,24)
		u ₁₁ (15,24)



Algorithm A*(u₀, Sol)

$G \leftarrow \{u_0\}$; $D \leftarrow \emptyset$; $g(u_0) \leftarrow 0$; $f(u_0) \leftarrow 0$; $u \leftarrow \text{premier}(G)$;

while ($G \neq \emptyset$) **et** ($u \notin T$) **do**

$G \leftarrow G \setminus \{u\}$; $D \leftarrow D \cup \{u\}$; $v \leftarrow \text{suivant}(u)$;

while (v est valide) **do**

if ($(v \notin (D \cup G))$ **ou** ($g(v) > g(u) + k(u, v)$)) **then**

$g(v) \leftarrow g(u) + k(u, v)$; $f(v) \leftarrow g(v) + h(v)$;

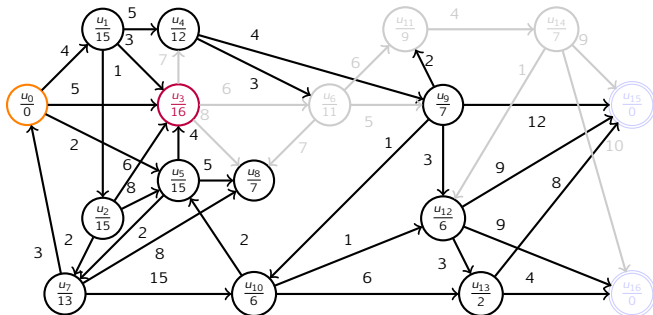
$\text{père}(v) \leftarrow u$; insérer-selon- $f(G, v)$;

$v \leftarrow \text{suivant}(u)$;

$u \leftarrow \text{premier}(G)$;

$v \leftarrow \perp$ $D = \{u_0, u_1, u_2, u_4, u_5, u_7, u_8, u_9, u_{10}, u_{12}, u_{13}\}$

$u(g(u), f(u))$	$\text{père}(u)$	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20)u_3(5,21)u_6(12,23)u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20)u_3(5,21)u_{12}(16,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{10}(14,20)$	u_9	$u_{12}(15,21)u_3(5,21)u_{13}(20,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{12}(15,21)$	u_{10}	$u_{13}(18,20)u_3(5,21)u_6(12,23)u_{15}(24,24)$
		$u_{16}(24,24)u_{11}(15,24)$
$u_{13}(18,20)$	u_{12}	$u_3(5,21)u_{16}(22,22)u_6(12,23)u_{15}(24,24)$
$u_3(5,21)$	u_0	$u_{11}(15,24)$



Algorithm A*(u₀, Sol)

$G \leftarrow \{u_0\}$; $D \leftarrow \emptyset$; $g(u_0) \leftarrow 0$; $f(u_0) \leftarrow 0$; $u \leftarrow \text{premier}(G)$;

while ($G \neq \emptyset$) **et** ($u \in T$) **do**

$G \leftarrow G \setminus \{u\}$; $D \leftarrow D \cup \{u\}$; $v \leftarrow \text{suivant}(u)$;

while (v est valide) **do**

if ($(v \notin (D \cup G))$ **ou** ($g(v) > g(u) + k(u, v)$)) **then**

$g(v) \leftarrow g(u) + k(u, v)$; $f(v) \leftarrow g(v) + h(v)$;

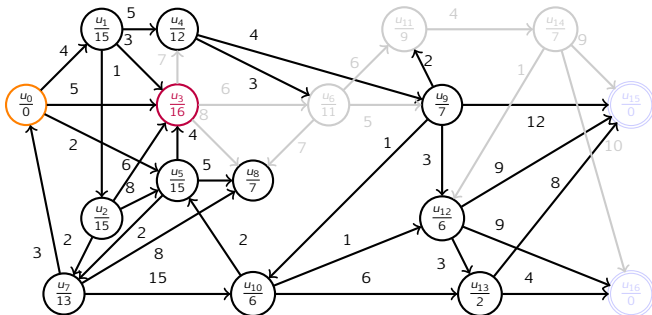
$\text{père}(v) \leftarrow u$; insérer-selon- $f(G, v)$;

$v \leftarrow \text{suivant}(u)$;

$u \leftarrow \text{premier}(G)$;

$v = \perp$ $D = \{u_0, u_1, u_2, u_4, u_5, u_7, u_8, u_9, u_{10}, u_{12}, u_{13}\}$

$u(g(u), f(u))$	$\text{père}(u)$	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20)u_3(5,21)u_6(12,23)u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20)u_3(5,21)u_{12}(16,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{10}(14,20)$	u_9	$u_{12}(15,21)u_3(5,21)u_{13}(20,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{12}(15,21)$	u_{10}	$u_{13}(18,20)u_3(5,21)u_6(12,23)u_{15}(24,24)$
		$u_{16}(24,24)u_{11}(15,24)$
$u_{13}(18,20)$	u_{12}	$u_3(5,21)u_{16}(22,22)u_6(12,23)u_{15}(24,24)$
		$u_{11}(15,24)$
$u_3(5,21)$	u_0	



Algorithme $A^*(u_0, \text{Sol})$

$G \leftarrow \{u_0\}$; $D \leftarrow \emptyset$; $g(u_0) \leftarrow 0$; $f(u_0) \leftarrow 0$; $u \leftarrow \text{premier}(G)$;

while ($G \neq \emptyset$) **et** ($u \notin T$) **do**

$G \leftarrow G \setminus \{u\}$; $D \leftarrow D \cup \{u\}$; $v \leftarrow \text{suivant}(u)$;

while (v est valide) **do**

if ($(v \notin (D \cup G))$ **ou** ($g(v) > g(u) + k(u, v)$)) **then**

$g(v) \leftarrow g(u) + k(u, v)$; $f(v) \leftarrow g(v) + h(v)$;

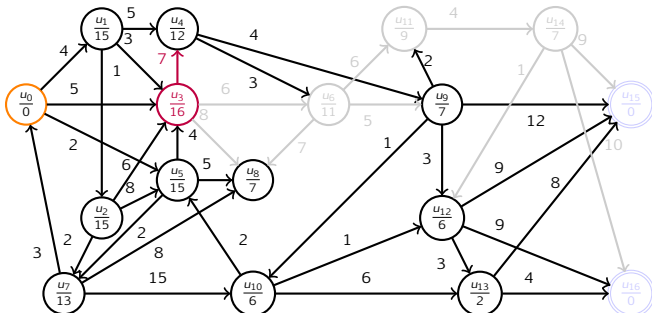
$\text{père}(v) \leftarrow u$; insérer-selon- $f(G, v)$;

$v \leftarrow \text{suivant}(u)$;

$u \leftarrow \text{premier}(G)$;

$v = u_4$ $D = \{u_0, u_1, u_2, u_3, u_4, u_5, u_7, u_8, u_9, u_{10}, u_{12}, u_{13}\}$

$u(g(u), f(u))$	$\text{père}(u)$	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20)u_3(5,21)u_6(12,23)u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20)u_3(5,21)u_{12}(16,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{10}(14,20)$	u_9	$u_{12}(15,21)u_3(5,21)u_{13}(20,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{12}(15,21)$	u_{10}	$u_{13}(18,20)u_3(5,21)u_6(12,23)u_{15}(24,24)$
		$u_{16}(24,24)u_{11}(15,24)$
$u_{13}(18,20)$	u_{12}	$u_3(5,21)u_{16}(22,22)u_6(12,23)u_{15}(24,24)$
		$u_{11}(15,24)$
$u_3(5,21)$	u_0	$u_{16}(22,22)u_6(12,23)u_{15}(24,24)$
		$u_{11}(15,24)$



Algorithme $A^*(u_0, \text{Sol})$

$G \leftarrow \{u_0\}$; $D \leftarrow \emptyset$; $g(u_0) \leftarrow 0$; $f(u_0) \leftarrow 0$; $u \leftarrow \text{premier}(G)$;

while ($G \neq \emptyset$) **et** ($u \notin T$) **do**

$G \leftarrow G \setminus \{u\}$; $D \leftarrow D \cup \{u\}$; $v \leftarrow \text{suivant}(u)$;

while (**v est valide**) **do**

if ($(v \notin (D \cup G))$ **ou** ($g(v) > g(u) + k(u, v)$)) **then**

$g(v) \leftarrow g(u) + k(u, v)$; $f(v) \leftarrow g(v) + h(v)$;

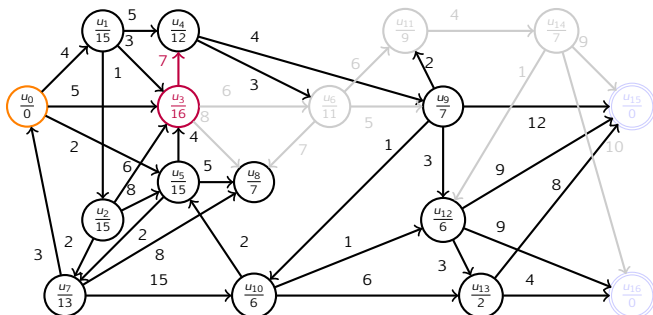
$\text{père}(v) \leftarrow u$; insérer-selon- $f(G, v)$;

$v \leftarrow \text{suivant}(u)$;

$u \leftarrow \text{premier}(G)$;

$v = u_4$ $D = \{u_0, u_1, u_2, u_3, u_4, u_5, u_7, u_8, u_9, u_{10}, u_{12}, u_{13}\}$

$u(g(u), f(u))$	$\text{père}(u)$	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20)u_3(5,21)u_6(12,23)u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20)u_3(5,21)u_{12}(16,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{10}(14,20)$	u_9	$u_{12}(15,21)u_3(5,21)u_{13}(20,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{12}(15,21)$	u_{10}	$u_{13}(18,20)u_3(5,21)u_6(12,23)u_{15}(24,24)$
		$u_{16}(24,24)u_{11}(15,24)$
$u_{13}(18,20)$	u_{12}	$u_3(5,21)u_{16}(22,22)u_6(12,23)u_{15}(24,24)$
		$u_{11}(15,24)$
$u_3(5,21)$	u_0	$u_{16}(22,22)u_6(12,23)u_{15}(24,24)$
		$u_{11}(15,24)$



Algorithme $A^*(u_0, \text{Sol})$

$$G \leftarrow \{u_0\}; D \leftarrow \emptyset; g(u_0) \leftarrow 0; f(u_0) \leftarrow 0; u \leftarrow \text{premier}(G);$$

```
while ( $G \neq \emptyset$ ) et ( $u \notin T$ ) do
```

$$G \leftarrow G \setminus \{u\}; D \leftarrow D \cup \{u\}; v \leftarrow \text{suivant}(u);$$

```
while (v est valide) do
```

if $((v \notin (D \cup G)) \text{ ou } (g(v) > g(u) + k(u,v)))$ then

$$g(v) \leftarrow g(u) + k(u,v); f(v) \leftarrow g(v) + h(v);$$

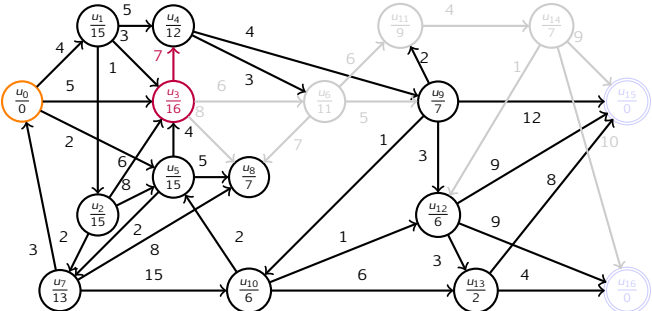
père(v) ← u ; insérer-selon-f(G,v);

```
v ← suivant(u);
```

```
u ← premier(G);
```

$$v=u_4 \quad D=\{u_0, u_1, u_2, u_3, u_4, u_5, u_7, u_8, u_9, u_{10}, u_{12}, u_{13}\}$$

$u(g(u), f(u))$	$père(u)$	$G = \{ v \mid v(g(v), f(v)) \}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20)u_3(5,21)u_6(12,23)u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20)u_3(5,21)u_{12}(16,22)u_6(12,23)$ $u_{11}(15,24)u_{15}(25,25)$
$u_{10}(14,20)$	u_9	$u_{12}(15,21)u_3(5,21)u_{13}(20,22)u_6(12,23)$ $u_{11}(15,24)u_{15}(25,25)$
$u_{12}(15,21)$	u_{10}	$u_{13}(18,20)u_3(5,21)u_6(12,23)u_{15}(24,24)$ $u_{16}(24,24)u_{11}(15,24)$
$u_{13}(18,20)$	u_{12}	$u_3(5,21)u_{16}(22,22)u_6(12,23)u_{15}(24,24)$ $u_{11}(15,24)$
$u_3(5,21)$	u_0	$u_{16}(22,22)u_6(12,23)u_{15}(24,24)$ $u_{11}(15,24)$



Algorithme $A^*(u_0, \text{Sol})$

$G \leftarrow \{u_0\}$; $D \leftarrow \emptyset$; $g(u_0) \leftarrow 0$; $f(u_0) \leftarrow 0$; $u \leftarrow \text{premier}(G)$;

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$g(v) \leftarrow g(u) + k(u, v)$; $f(v) \leftarrow g(v) + h(v)$;

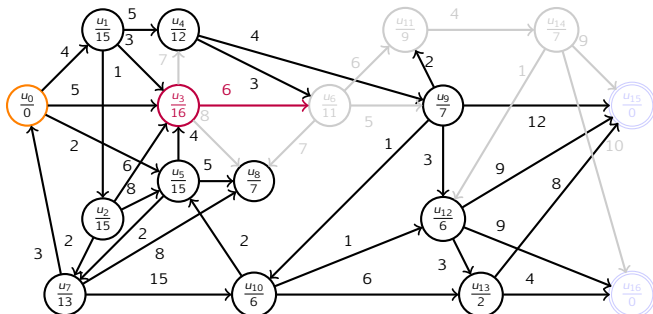
$\text{père}(v) \leftarrow u$; insérer-selon- $f(G, v)$;

$v \leftarrow \text{suivant}(u)$;

$u \leftarrow \text{premier}(G)$;

$v = u_6$ $D = \{u_0, u_1, u_2, u_3, u_4, u_5, u_7, u_8, u_9, u_{10}, u_{12}, u_{13}\}$

$u(g(u), f(u))$	$\text{père}(u)$	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20)u_3(5,21)u_6(12,23)u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20)u_3(5,21)u_{12}(16,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{10}(14,20)$	u_9	$u_{12}(15,21)u_3(5,21)u_{13}(20,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{12}(15,21)$	u_{10}	$u_{13}(18,20)u_3(5,21)u_6(12,23)u_{15}(24,24)$
		$u_{16}(24,24)u_{11}(15,24)$
$u_{13}(18,20)$	u_{12}	$u_3(5,21)u_{16}(22,22)u_6(12,23)u_{15}(24,24)$
		$u_{11}(15,24)$
$u_3(5,21)$	u_0	$u_{16}(22,22)u_6(12,23)u_{15}(24,24)$
		$u_{11}(15,24)$



Algorithme $A^*(u_0, \text{Sol})$

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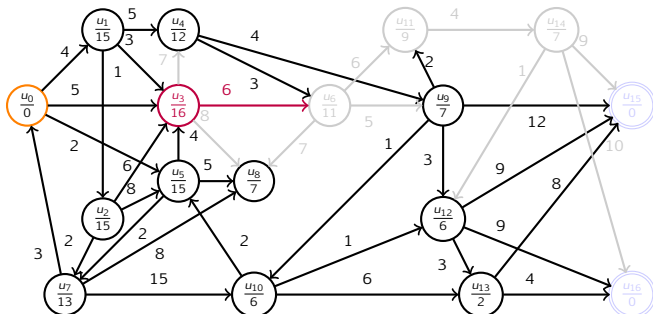
$\text{père}(v) \leftarrow u$; $\text{insérer-selon-}f(G, v)$;

$v \leftarrow \text{suivant}(u)$;

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$v = u_6$ $D = \{u_0, u_1, u_2, u_3, u_4, u_5, u_7, u_8, u_9, u_{10}, u_{12}, u_{13}\}$

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$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20)u_3(5,21)u_6(12,23)u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20)u_3(5,21)u_{12}(16,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{10}(14,20)$	u_9	$u_{12}(15,21)u_3(5,21)u_{13}(20,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{12}(15,21)$	u_{10}	$u_{13}(18,20)u_3(5,21)u_6(12,23)u_{15}(24,24)$
		$u_{16}(24,24)u_{11}(15,24)$
$u_{13}(18,20)$	u_{12}	$u_3(5,21)u_{16}(22,22)u_6(12,23)u_{15}(24,24)$
		$u_{11}(15,24)$
$u_3(5,21)$	u_0	$u_{16}(22,22)u_6(12,23)u_{15}(24,24)$
		$u_{11}(15,24)$



Algorithme $A^*(u_0, \text{Sol})$

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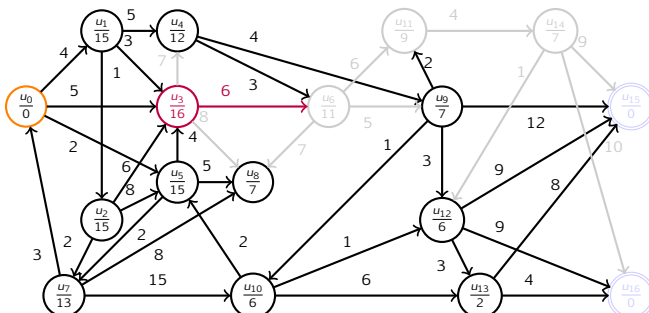
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$v \leftarrow \text{suivant}(u)$;

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$u(g(u), f(u))$	$\text{père}(u)$	$G = \{v \mid v(g(v), f(v))\}$
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$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20)u_3(5,21)u_6(12,23)u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20)u_3(5,21)u_{12}(16,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{10}(14,20)$	u_9	$u_{12}(15,21)u_3(5,21)u_{13}(20,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{12}(15,21)$	u_{10}	$u_{13}(18,20)u_3(5,21)u_6(12,23)u_{15}(24,24)$
		$u_{16}(24,24)u_{11}(15,24)$
$u_{13}(18,20)$	u_{12}	$u_3(5,21)u_{16}(22,22)u_6(12,23)u_{15}(24,24)$
		$u_{11}(15,24)$
$u_3(5,21)$	u_0	$u_{16}(22,22)u_6(12,23)u_{15}(24,24)$
		$u_{11}(15,24)$



Algorithme $A^*(u_0, \text{Sol})$

$G \leftarrow \{u_0\}$; $D \leftarrow \emptyset$; $g(u_0) \leftarrow 0$; $f(u_0) \leftarrow 0$; $u \leftarrow \text{premier}(G)$;

while ($G \neq \emptyset$) **et** ($u \notin T$) **do**

$G \leftarrow G \setminus \{u\}$; $D \leftarrow D \cup \{u\}$; $v \leftarrow \text{suivant}(u)$;

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$g(v) \leftarrow g(u) + k(u, v)$; $f(v) \leftarrow g(v) + h(v)$;

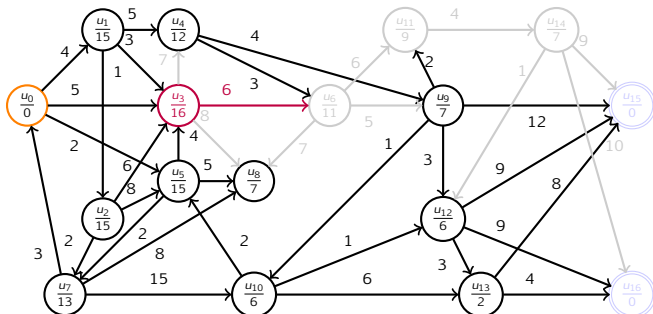
$\text{père}(v) \leftarrow u$; $\text{insérer-selon-}f(G, v)$;

$v \leftarrow \text{suivant}(u)$;

$u \leftarrow \text{premier}(G)$;

$v = u_6$ $D = \{u_0, u_1, u_2, u_3, u_4, u_5, u_7, u_8, u_9, u_{10}, u_{12}, u_{13}\}$

$u(g(u), f(u))$	$\text{père}(u)$	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20)u_3(5,21)u_6(12,23)u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20)u_3(5,21)u_{12}(16,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{10}(14,20)$	u_9	$u_{12}(15,21)u_3(5,21)u_{13}(20,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{12}(15,21)$	u_{10}	$u_{13}(18,20)u_3(5,21)u_6(12,23)u_{15}(24,24)$
		$u_{16}(24,24)u_{11}(15,24)$
$u_{13}(18,20)$	u_{12}	$u_3(5,21)u_{16}(22,22)u_6(12,23)u_{15}(24,24)$
		$u_{11}(15,24)$
$u_3(5,21)$	u_0	$u_{16}(22,22)u_6(11,22)u_{15}(24,24)$
		$u_{11}(15,24)$



Algorithm A*(u₀, Sol)

$G \leftarrow \{u_0\}$; $D \leftarrow \emptyset$; $g(u_0) \leftarrow 0$; $f(u_0) \leftarrow 0$; $u \leftarrow \text{premier}(G)$;

while ($G \neq \emptyset$) **et** ($u \notin T$) **do**

$G \leftarrow G \setminus \{u\}$; $D \leftarrow D \cup \{u\}$; $v \leftarrow \text{suivant}(u)$;

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if ($(v \notin (D \cup G))$ **ou** ($g(v) > g(u) + k(u, v)$)) **then**

$g(v) \leftarrow g(u) + k(u, v)$; $f(v) \leftarrow g(v) + h(v)$;

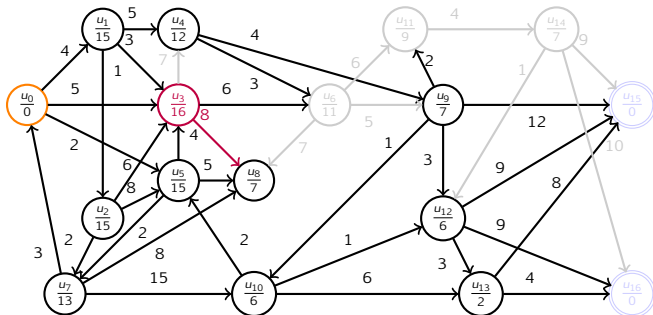
$\text{père}(v) \leftarrow u$; insérer-selon- $f(G, v)$;

$v \leftarrow \text{suivant}(u)$;

$u \leftarrow \text{premier}(G)$;

$v = u_8$ $D = \{u_0, u_1, u_2, u_3, u_4, u_5, u_7, u_8, u_9, u_{10}, u_{12}, u_{13}\}$

$u(g(u), f(u))$	$\text{père}(u)$	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
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$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20)u_3(5,21)u_6(12,23)u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20)u_3(5,21)u_{12}(16,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{10}(14,20)$	u_9	$u_{12}(15,21)u_3(5,21)u_{13}(20,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{12}(15,21)$	u_{10}	$u_{13}(18,20)u_3(5,21)u_6(12,23)u_{15}(24,24)$
		$u_{16}(24,24)u_{11}(15,24)$
$u_{13}(18,20)$	u_{12}	$u_3(5,21)u_{16}(22,22)u_6(12,23)u_{15}(24,24)$
		$u_{11}(15,24)$
$u_3(5,21)$	u_0	$u_{16}(22,22)u_6(11,22)u_{15}(24,24)u_{11}(15,24)$



Algorithm A*(u₀, Sol)

$G \leftarrow \{u_0\}$; $D \leftarrow \emptyset$; $g(u_0) \leftarrow 0$; $f(u_0) \leftarrow 0$; $u \leftarrow \text{premier}(G)$;

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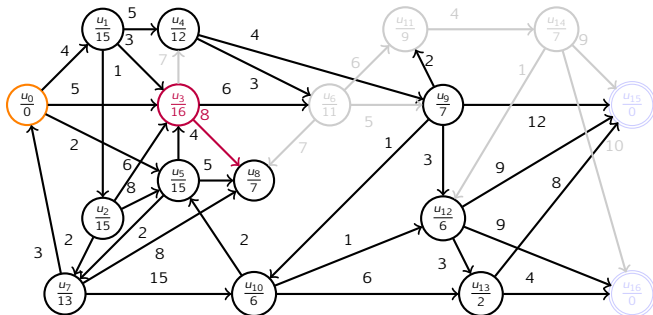
$\text{père}(v) \leftarrow u$; insérer-selon- $f(G, v)$;

$v \leftarrow \text{suivant}(u)$;

$u \leftarrow \text{premier}(G)$;

$v = u_8$ $D = \{u_0, u_1, u_2, u_3, u_4, u_5, u_7, u_8, u_9, u_{10}, u_{12}, u_{13}\}$

$u(g(u), f(u))$	$\text{père}(u)$	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
$u_8(7,14)$	u_5	$u_7(4,17)u_1(4,19)u_3(5,21)$
$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20)u_3(5,21)u_6(12,23)u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20)u_3(5,21)u_{12}(16,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{10}(14,20)$	u_9	$u_{12}(15,21)u_3(5,21)u_{13}(20,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{12}(15,21)$	u_{10}	$u_{13}(18,20)u_3(5,21)u_6(12,23)u_{15}(24,24)$
		$u_{16}(24,24)u_{11}(15,24)$
$u_{13}(18,20)$	u_{12}	$u_3(5,21)u_{16}(22,22)u_6(12,23)u_{15}(24,24)$
		$u_{11}(15,24)$
$u_3(5,21)$	u_0	$u_{16}(22,22)u_6(11,22)u_{15}(24,24)u_{11}(15,24)$



Algorithm A*(u₀, Sol)

$G \leftarrow \{u_0\}$; $D \leftarrow \emptyset$; $g(u_0) \leftarrow 0$; $f(u_0) \leftarrow 0$; $u \leftarrow \text{premier}(G)$;

while ($G \neq \emptyset$) **et** ($u \notin T$) **do**

$G \leftarrow G \setminus \{u\}$; $D \leftarrow D \cup \{u\}$; $v \leftarrow \text{suivant}(u)$;

while (v est valide) **do**

if ($((v \in (D \cup G)) \text{ ou } (g(v) > g(u) + k(u,v)))$ **then**

$g(v) \leftarrow g(u) + k(u,v)$; $f(v) \leftarrow g(v) + h(v)$;

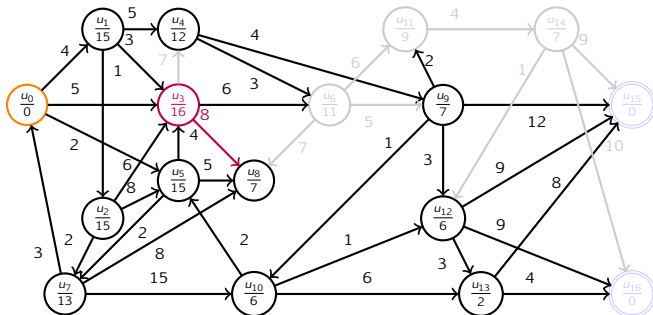
$\text{père}(v) \leftarrow u$; insérer-selon- $f(G,v)$;

$v \leftarrow \text{suivant}(u)$;

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$v = u_8$ $D = \{u_0, u_1, u_2, u_3, u_4, u_5, u_7, u_8, u_9, u_{10}, u_{12}, u_{13}\}$

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$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
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$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20)u_3(5,21)u_6(12,23)u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20)u_3(5,21)u_{12}(16,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{10}(14,20)$	u_9	$u_{12}(15,21)u_3(5,21)u_{13}(20,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{12}(15,21)$	u_{10}	$u_{13}(18,20)u_3(5,21)u_6(12,23)u_{15}(24,24)$
		$u_{16}(24,24)u_{11}(15,24)$
$u_{13}(18,20)$	u_{12}	$u_3(5,21)u_{16}(22,22)u_6(12,23)u_{15}(24,24)$
		$u_{11}(15,24)$
$u_3(5,21)$	u_0	$u_{16}(22,22)u_6(11,22)u_{15}(24,24)u_{11}(15,24)$



Algorithm A*(u₀, Sol)

$G \leftarrow \{u_0\}$; $D \leftarrow \emptyset$; $g(u_0) \leftarrow 0$; $f(u_0) \leftarrow 0$; $u \leftarrow \text{premier}(G)$;

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$G \leftarrow G \setminus \{u\}$; $D \leftarrow D \cup \{u\}$; $v \leftarrow \text{suivant}(u)$;

while (v est valide) **do**

if ($(v \notin (D \cup G))$ **ou** ($g(v) > g(u) + k(u,v)$)) **then**

$g(v) \leftarrow g(u) + k(u,v)$; $f(v) \leftarrow g(v) + h(v)$;

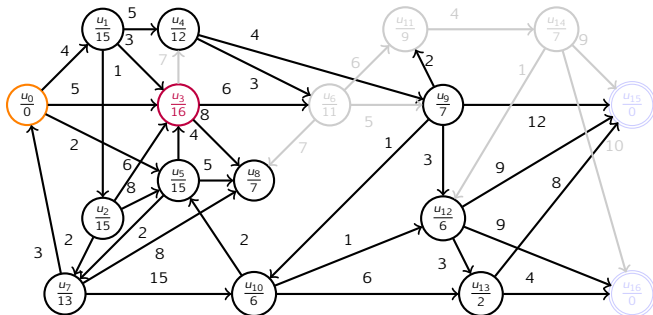
$\text{père}(v) \leftarrow u$; insérer-selon- $f(G,v)$;

$v \leftarrow \text{suivant}(u)$;

$u \leftarrow \text{premier}(G)$;

$v \leftarrow \perp$; $D = \{u_0, u_1, u_2, u_3, u_4, u_5, u_7, u_8, u_9, u_{10}, u_{12}, u_{13}\}$

$u(g(u), f(u))$	$\text{père}(u)$	$G = \{v \mid v(g(v), f(v))\}$
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$u_7(4,17)$	u_5	$u_1(4,19)u_3(5,21)u_{10}(19,25)$
$u_1(4,19)$	u_0	$u_2(5,20)u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_2(5,20)$	u_1	$u_4(9,21)u_3(5,21)u_{10}(19,25)$
$u_4(9,21)$	u_1	$u_9(13,20)u_3(5,21)u_6(12,23)u_{10}(19,25)$
$u_9(13,20)$	u_4	$u_{10}(14,20)u_3(5,21)u_{12}(16,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{10}(14,20)$	u_9	$u_{12}(15,21)u_3(5,21)u_{13}(20,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{12}(15,21)$	u_{10}	$u_{13}(18,20)u_3(5,21)u_6(12,23)u_{15}(24,24)$
		$u_{16}(24,24)u_{11}(15,24)$
$u_{13}(18,20)$	u_{12}	$u_3(5,21)u_{16}(22,22)u_6(12,23)u_{15}(24,24)$
		$u_{11}(15,24)$
$u_3(5,21)$	u_0	$u_{16}(22,22)u_6(11,22)u_{15}(24,24)u_{11}(15,24)$



Algorithm A*(u₀, Sol)

$G \leftarrow \{u_0\}$; $D \leftarrow \emptyset$; $g(u_0) \leftarrow 0$; $f(u_0) \leftarrow 0$; $u \leftarrow \text{premier}(G)$;

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while (**v est valide**) **do**

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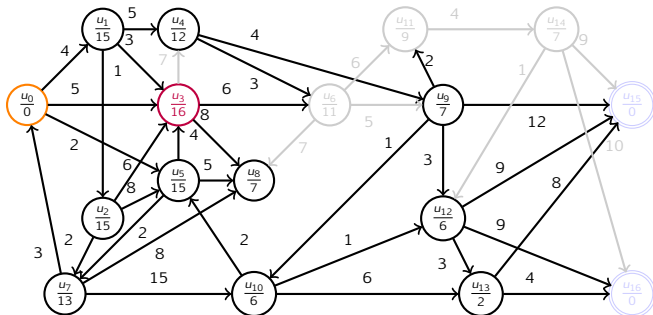
$\text{père}(v) \leftarrow u$; insérer-selon- $f(G, v)$;

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		$u_{11}(15,24)u_{15}(25,25)$
$u_{10}(14,20)$	u_9	$u_{12}(15,21)u_3(5,21)u_{13}(20,22)u_6(12,23)$
		$u_{11}(15,24)u_{15}(25,25)$
$u_{12}(15,21)$	u_{10}	$u_{13}(18,20)u_3(5,21)u_6(12,23)u_{15}(24,24)$
		$u_{16}(24,24)u_{11}(15,24)$
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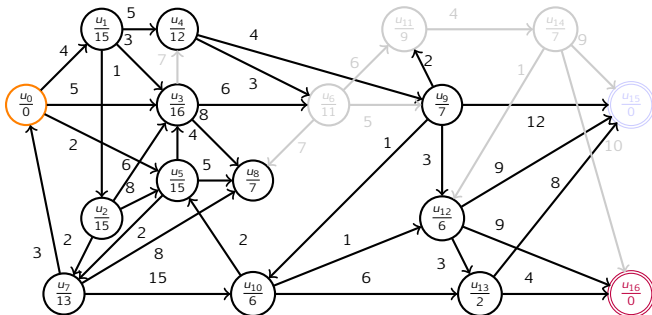
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$u_{12}(15,21)$	u_{10}	$u_{13}(18,20)u_3(5,21)u_6(12,23)u_{15}(24,24)$
		$u_{16}(24,24)u_{11}(15,24)$
$u_{13}(18,20)$	u_{12}	$u_3(5,21)u_{16}(22,22)u_6(12,23)u_{15}(24,24)$
		$u_{11}(15,24)$
$u_3(5,21)$	u_0	$u_{16}(22,22)u_6(11,22)u_{15}(24,24)u_{11}(15,24)$
$u_{16}(22,22)$	u_{13}	



Algorithm A*(u₀, Sol)

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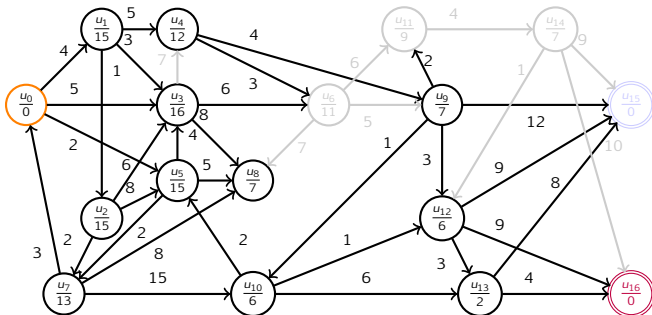
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$u_9(13,20)$	u_4	$u_{10}(14,20)u_3(5,21)u_{12}(16,22)u_6(12,23)$
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$u_{10}(14,20)$	u_9	$u_{12}(15,21)u_3(5,21)u_{13}(20,22)u_6(12,23)$
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$u_{12}(15,21)$	u_{10}	$u_{13}(18,20)u_3(5,21)u_6(12,23)u_{15}(24,24)$
		$u_{16}(24,24)u_{11}(15,24)$
$u_{13}(18,20)$	u_{12}	$u_3(5,21)u_{16}(22,22)u_6(12,23)u_{15}(24,24)$
		$u_{11}(15,24)$
$u_3(5,21)$	u_0	$u_{16}(22,22)u_6(11,22)u_{15}(24,24)u_{11}(15,24)$
$u_{16}(22,22)$	u_{13}	



Algorithm A*(u₀, Sol)

```

G ← {u0}; D ← ∅; g(u0) ← 0; f(u0) ← 0; u ← premier(G);
while (G ≠ ∅) et (u ∈ T) do
  G ← G \ {u}; D ← D ∪ {u}; v ← suivant(u);
  while (v est valide) do
    if ((v ∉ (D ∪ G)) ou (g(v) > g(u) + k(u,v))) then
      g(v) ← g(u) + k(u,v); f(v) ← g(v) + h(v);
      père(v) ← u; insérer-selon-f(G,v);
    v ← suivant(u);
  u ← premier(G);

```

$u(g(u), f(u))$	père(u)	$G = \{v \mid v(g(v), f(v))\}$
$u_0(0,0)$	$\perp$	$u_5(2,17)u_1(4,19)u_3(5,21)$
$u_5(2,17)$	u_0	$u_8(7,14)u_7(4,17)u_1(4,19)u_3(5,21)$
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		$u_{11}(15,24)$
$u_3(5,21)$	u_0	$u_{16}(22,22)u_6(11,22)u_{15}(24,24)u_{11}(15,24)$
$u_{16}(22,22)$	u_{13}	

La solution est {u₀, u₁, u₄, u₉, u₁₀, u₁₂, u₁₃, u₁₆} avec un coût de 22 (=f(u₁₆))

